Bachelor’s thesis

HMM for HFT

Predicting and trading on short-term regimes in currency tick data

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Abstract

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1 Introduction

1.1 The quantitative revolution

Modern financial markets are increasingly mathematical objects. What was once organized around human brokers, trading pits, voice execution, and discretionary judgement has gradually become an ecosystem of electronic limit-order books, automated market makers, statistical arbitrage engines, and latency-sensitive execution algorithms. This transformation is often described through the rise of high-frequency trading, but its deeper significance is the replacement of narrative-based market interpretation with probabilistic modelling, signal extraction, optimization, and real-time statistical decision-making. In this setting, prices are not merely economic summaries of fundamentals; they are also high-dimensional data streams generated by the interaction of heterogeneous agents, algorithms, inventories, order flows, and information shocks.

The popular account of this transition often begins with the drama of speed: faster cables, co-location, microwave links, and the race to react to information milliseconds before competitors (Lewis 2014; Aldridge 2013). However, from a mathematical perspective, speed is only one dimension of the problem. The more fundamental issue is that financial markets produce noisy, non-stationary time series in which any exploitable structure is weak, transient, and rapidly arbitraged away. This makes quantitative trading a natural domain for applied probability, statistics, stochastic processes, and machine learning. The trader is no longer simply asking whether an asset is "cheap" or "expensive", but whether a small statistical regularity can be detected, validated, traded, and risk-managed after costs.

A useful symbolic figure in this quantitative revolution is Jim Simons, whose background in mathematics and codebreaking helped define the modern idea of finance as a signal-processing problem rather than a purely economic forecasting problem. The lesson is not that markets are deterministic or easily predictable, but almost the opposite: if markets are noisy, adaptive, and adversarial, then any predictive edge must be discovered through large-scale data analysis, rigorous out-of-sample testing, and careful modelling of uncertainty. This philosophy is closely related to the theme of this project. Instead of trying to forecast long-run currency fundamentals, we study whether short-term statistical structure in currency tick data can be captured by models that explicitly allow the market environment to change through time.

1.2 The microstructure of FX and market makers

The foreign exchange market is one of the most natural arenas in which to study this problem. FX is globally traded, highly liquid, electronically intermediated, and active nearly twenty-four hours a day. At the daily or weekly horizon, exchange rates are often modelled as approximately continuous stochastic processes driven by macroeconomic information, interest-rate expectations, risk sentiment, and international capital flows.

At the tick level, however, the market looks very different. Prices move in small discrete increments; bid and ask quotes update asynchronously; liquidity appears and disappears; and the observed price path becomes a sequence of small jumps, reversals, and microstructural frictions rather than a smooth curve.

This microscopic view is the domain of market microstructure. The central question is no longer only what the “true” value of a currency is, but how trading mechanisms, inventory constraints, asymmetric information, and order flow jointly determine observed prices (O’Hara 1995; Hasbrouck 2007). A market maker, for example, typically provides liquidity by quoting both bid and ask prices, earning compensation through the spread while managing inventory and adverse-selection risk. In a quiet regime, the market maker may be able to provide liquidity profitably as prices oscillate locally. In a stressed or information-driven regime, however, the same strategy may become dangerous: order flow can become one-sided, spreads may widen, and price changes may become directional rather than mean-reverting.

This distinction is crucial for high-frequency modelling. At lower frequencies, Brownian motion and diffusion models offer a powerful first approximation to asset-price dynamics, and the connection between finance and physics becomes visible through random walks, diffusion equations, and stochastic calculus (Bachelier 1900; Black and Scholes 1973; Merton 1973). At the tick level, however, the idealized diffusion picture becomes contaminated by the machinery of trading itself. Bid–ask bounce, discreteness, latency, varying liquidity, and order-book dynamics create patterns that are not well captured by a single stationary Gaussian model. The empirical object studied in this thesis is therefore not a clean theoretical Brownian path, but a noisy microstructural signal in which the statistical properties of returns may change rapidly over time.

Currency pairs are especially interesting in this regard because they often have economic and geographical links. Pairs such as AUD/USD and NZD/USD are exposed to related regional and commodity-market forces; EUR/NOK and EUR/SEK connect two Nordic currencies against the euro; and GBP/USD and EUR/USD share the US dollar leg and are deeply embedded in global macro trading. These relationships make currency pairs natural candidates for statistical arbitrage and pairs trading: rather than taking an outright directional view on a single exchange rate, one can model the relative movement between two related exchange rates and test whether temporary deviations tend to revert.

1.3 Non-stationarity and market regimes

The main statistical difficulty is non-stationarity. A standard time-series model usually assumes that the same parameters govern the full sample, or at least that the estimated relationship is stable enough to remain useful out of sample. In financial markets this assumption is often too strong. Volatility clusters, correlations rise and fall, liquidity varies intraday, and the same trading rule may behave very differently depending on whether the market is calm, trending, news-driven, or illiquid. This is not merely an inconvenience; it is one of the defining mathematical features of financial data (Engle 1982; Cont 2001).

For pairs trading, non-stationarity is particularly damaging. A spread that appears mean-reverting in one period may drift in another. A hedge ratio estimated on a long historical sample may be irrelevant over a shorter local window. A pair may be statistically related in the full sample but too slow-moving to trade after transaction costs, or it may exhibit brief episodes of local mean reversion that are invisible to a static model. Thus, the empirical challenge is not only to detect dependence between two currency pairs, but to determine when this dependence is strong enough, stable enough, and cheap enough to trade.

This motivates the language of market regimes. Informally, a regime is a persistent market state with its own statistical properties. One regime may be characterized by low volatility, tight spreads, and local mean reversion; another by high volatility, weak pair-dependence, and directional price movement. The regime itself is not directly observed. We observe only the emitted data: returns, spreads, rolling volatility, correlations, and trading outcomes. The modelling problem is therefore naturally probabilistic: infer the hidden state from the observable time series, and use that inference to adapt the trading rule.

1.4 The solution in Hidden Markov Models

Hidden Markov Models provide a mathematically structured way to represent this idea. The model assumes that the observed process is driven by an unobserved Markov chain whose state determines the distribution of returns. The Markov assumption is deliberately simple: conditional on the current hidden state, the probability of the next state depends only on the present state and not on the full past. Despite this simplicity, the framework is powerful because it allows persistence, sudden switching, and regime-dependent parameters within a single probabilistic model (Rabiner 1989; Hamilton 1989).

In this project, the Hidden Markov Model is used as a regime filter for high-frequency currency-pair trading. The baseline model captures simple spread-based trading behaviour, while the autoregressive extension allows short-term serial dependence to enter the signal. The Markov-switching autoregressive model then allows the parameters of the return process to depend on the inferred market regime. Intuitively, the model asks whether there are hidden "safe" and "dangerous" states in the tick data, and whether avoiding the dangerous state improves the economic performance of a pairs-trading strategy.

This approach is attractive because it is interpretable. Unlike a black-box classifier, the HMM produces transition probabilities, regime-specific volatilities, regime-specific expected returns, and smoothed state probabilities. These quantities can be inspected directly and connected to financial intuition: a high-variance regime may correspond to unstable market conditions, while a low-variance regime may correspond to a more favourable environment for mean reversion. In this sense, the HMM is not only a predictive model but also a diagnostic tool for studying how microstructural conditions evolve through time.

At the same time, the project is deliberately empirical and sceptical. The purpose is not to assume that a more complex model must outperform a simpler one. In high-frequency finance, additional model complexity can easily overfit noise, especially when trading costs, bid–ask spreads, and walk-forward validation are taken seriously. The key question is therefore comparative: does a Hidden Markov regime filter add value relative to simpler baseline and autoregressive trading rules, or does it mainly reduce exposure without improving risk-adjusted performance?

1.5 Thesis structure

The remainder of the thesis is organized as follows. Section 2 introduces the statistical and financial methodology. It begins with the autoregressive benchmark model, then develops the Hidden Markov and Markov-switching autoregressive framework, before describing the data construction, model-selection procedure, transaction-cost assumptions, and backtesting metrics. Particular emphasis is placed on evaluating performance net of realistic trading frictions, since high-frequency strategies are highly sensitive to spreads and turnover.

Section 3 presents the empirical results for three currency-pair case studies: AUD/USD versus NZD/USD, EUR/NOK versus EUR/SEK, and GBP/USD versus EUR/USD. For each pair, the analysis studies liquidity, return distributions, pair-dependence, cointegration diagnostics, walk-forward optimization, and strategy performance. The baseline, autoregressive, and Markov-switching autoregressive strategies are compared in terms of total return, volatility, drawdown, Sharpe ratio, exposure, win rate, and tail behaviour.

Section 4 concludes by evaluating whether Hidden Markov Models are useful for high-frequency FX pairs trading. The central contribution of the thesis is not a claim that HMMs universally outperform simpler models, but a careful test of when regime awareness helps, when it fails, and how mathematical models behave when confronted with noisy, costly, non-stationary tick data.

2 Methods

In this section I introduce the methods. Namely, we start by introducing autoregressive models, before writing about Hidden Markov Models (HMM), introducing the data used in our simulations, while also presenting the statistical and financial metrics used to score the performance of our trading simulations.

2.1 Autoregressive models

When modeling time series data, a fundamental objective is to capture the linear dependence between current and past observations. The general Autoregressive model of order p, denoted as AR(p), formalizes this intuition by projecting the current value of a variable Y_t as a linear combination of its p most recent past values:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t \tag{1}$$

In the context of financial returns, we establish our benchmark using the simplest form of this process: a single-regime autoregressive model of order one, or AR(1). This model serves as a robust baseline because it efficiently captures short-term serial dependence, such as market momentum or immediate mean-reversion. The AR(1) process for returns r_t is defined as:

$$r_t = c + \phi r_{t-1} + \epsilon_t \tag{2}$$

Here, c is the constant intercept (from which the unconditional mean return can be derived as $\frac{c}{1-\phi}$), ϕ measures the strength and direction of the linear serial dependence, and ϵ_t represents the stochastic innovation term, strictly assumed to be independent and identically distributed as Gaussian white noise, where $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$.

2.2 Hidden Markov Models

While the standard AR(1) model provides a solid baseline, its primary limitation is the assumption of structural stability; it assumes that the parameters (c, ϕ, and σ^2) remain constant over time. Real-world financial markets, however, frequently undergo structural breaks and shift between distinct 'regimes'—such as transitioning from a low-volatility bull market to a high-volatility bear market.

To capture this dynamic behaviour, we extend our framework using a Hidden Markov Model (HMM). We introduce an unobserved, discrete-time Markov chain, denoted by $S_t \in \{1, \ldots, K\}$, which represents the active market regime at time t. The probability of the market transitioning from one state to another is governed by a set of transition probabilities:

$$\mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij} \tag{3}$$

where $p_{ij} \geq 0$ and the probabilities of transitioning from any state i to all possible states j sum to one.

$$\sum_{j=1}^{K} p_{ij} = 1 \quad \forall i \tag{4}$$

By conditioning our autoregressive process on this hidden state $S_t = k$, we allow the behavior of the returns to adapt to the current market environment. The regime-switching AR(1) model is therefore defined as:

$$r_t = \mu_k + \phi_k r_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_k^2) \tag{5}$$

This formulation is significantly more flexible than the single-regime benchmark. The parameter μ_k acts as the regime-specific drift, ϕ_k captures how the short-term memory of the market changes depending on the environment, and σ_k^2 models regime-specific volatility, effectively accommodating the volatility clustering frequently observed in financial time series.

To fully specify this Hidden Markov Model, we must estimate the complete parameter set θ, which encapsulates both the state-transition dynamics and the regime-dependent autoregressive parameters:

$$\theta = \{\pi, \mathbf{P}, \mu_k, \phi_k, \sigma_k^2 : k = 1, \ldots, K\} \tag{6}$$

where $\pi = (\pi_1, \ldots, \pi_K)^\top$ is the initial state distribution representing the probability $\mathbb{P}(S_1 = k)$ of starting in state k, $\mathbf{P} = (p_{ij})$ is the $K \times K$ transition matrix dictating the regime persistence, and the remaining terms represent the localized AR parameters for each respective state.

2.3 Model selection

After defining the structural framework of the Hidden Markov Model, the immediate empirical challenge is determining the optimal hyperparameters—most notably, the number of hidden states, K, and the autoregressive order, p. Adding more states or lags will invariably increase the model's fit to the training data, but doing so introduces the severe risk of overfitting, where the model captures statistical noise rather than genuine market structure. To resolve this tradeoff, we evaluate competing models using Information Criteria, specifically the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). Both metrics penalize models for unnecessary complexity while rewarding goodness-of-fit.

The AIC focuses on minimizing the information lost when a given model is used to approximate the true data-generating process, making it highly effective for optimizing predictive accuracy. It evaluates models based on m, which represents the total number of estimated parameters in the model, and $\hat{L}$, which is the maximized value of the likelihood function for the given data. The AIC is defined as:

$$\text{AIC} = 2m - 2\ln(\hat{L}) \tag{7}$$

The BIC introduces a stricter penalty for model complexity by factoring in T, the total number of observations in the time series. Because the penalty term $\ln(T)$ scales with the sample size, the BIC aggressively penalizes overfitting in large financial datasets and mathematically favors the most parsimonious model that adequately explains the data. The BIC is defined as:

$$\text{BIC} = m\ln(T) - 2\ln(\hat{L}) \tag{8}$$

During our hyperparameter sweep, models are estimated across a grid of potential K and p values. The configuration that minimizes the AIC and BIC is selected as the optimal structural architecture for our subsequent trading simulations.

2.4 Metrics

To rigorously evaluate the performance of the trading strategies generated in this simulation, we must rely on a robust set of quantitative metrics. Simply observing the final account balance is insufficient; a proper evaluation requires us to understand the path taken to achieve those returns, the risks assumed along the way, and the underlying mathematical behavior of the system. In this section, I introduce the comprehensive suite of metrics used to score the models. These indicators allow us to objectively compare different strategies, identifying their specific structural strengths, potential vulnerabilities, and overall viability in a live market context. Before detailing the specific calculations, we define the following standard notations used throughout this section:

- R_t: Return of the asset/strategy at time t
- μ: Mean return $\mathbb{E}(R)$
- σ Standard deviation of returns
- N: Total number of periods or trades
- V_t: Portfolio value at time t, and $P_t = \max_{s \le t} V_s$ is the peak value up to time t

2.4.1 Statistical metrics

Statistical metrics form the foundational mathematical profile of our trading strategies. Rather than focusing directly on capital accumulation, these indicators analyze the shape, spread, and probability distribution of the underlying return data. By examining how returns behave on a probabilistic level, we can uncover hidden risks that standard equity curves might obscure—such as the likelihood of extreme market anomalies or structural asymmetries in the strategy's outcomes. The following metrics are utilized to quantify the fundamental risk and distributional nature of the strategy:

Annual volatility quantifies the expected yearly fluctuation in returns, providing a baseline mathematical measure of standard risk.

$$\sigma_{\text{ann}} = \sigma\sqrt{T} \tag{9}$$

Skewness measures the assymetry of the return distribution, helping to identify the likelihood of outsized positive or negative outliers.

$$S = \frac{E[(R-\mu)^3]}{\sigma^3} \tag{10}$$

Kurtosis measures the 'fatness' of the distribution tails, indicating the probability of rare, extreme market events.

$$K = \frac{E[(R-\mu)^4]}{\sigma^4} \tag{11}$$

Value at risk estimates the maximum expected loss over a specific time frame at a given confidence level α, acting as a standard gauge of the worst-case scenario.

$$\mathrm{VaR}_\alpha = -Q_{1-\alpha}(R) \tag{12}$$

Conditional value at risk calculates the average expected loss assuming the VaR threshold is breached, providing a more comprehensive picture of extreme tail risk.

$$\mathrm{CVaR}_\alpha = -E[R \mid R \leq -\mathrm{VaR}_\alpha] \tag{13}$$

Tail factor quantifies the severy of tail risk by comparing the average extreme loss to the baseline extreme loss threshold.

$$\mathrm{TF} = \frac{\mathrm{CVaR}_\alpha}{\mathrm{VaR}_\alpha} \tag{14}$$

2.4.2 Financial metrics

While statistical metrics describe the mathematical probability of returns, financial metrics translate those probabilities into practical, capital-based outcomes. These indicators evaluate the actual equity curve, measuring how efficiently a strategy deploys capital, how it navigates periods of historical loss, and its absolute trading profitability. Crucially, these metrics contextualize raw returns against the specific drawdowns and volatility endured to achieve them, providing a holistic view of the strategy's execution quality and the psychological 'holding stress' it would impose on an investor. The core financial scoring metrics used in our evaluation are defined as follows:

Total return represents the overall cumulative growth or decline of the invested capital over the entire trading period.

$$\mathrm{TR} = \prod_{t=1}^{N} (1 + R_t) - 1 \tag{15}$$

Annual return standardises the cumulative return into a yearly figure, allowing for direct comparison across different strategies and timeframes.

$$\mathrm{AR} = (1 + \mathrm{TR})^{\frac{1}{Y}} - 1 \tag{16}$$

Positive months counts the total number of profitable months, offering a straightforward measure of performance consistency over time.

$$\mathrm{N}_{\mathrm{pos}} = \sum_{t=1}^{N} \mathbb{I}(R_t > 0) \tag{17}$$

Max drawdown measures the largest peak-to-trough percentage drop in equity, revealing the strategy's worst historical losing streak.

$$\mathrm{MDD} = \max_t \left(\frac{P_t - V_t}{P_t} \right) \tag{18}$$

Max drawdown duration measures the longest time required to recover from a peak-to-trough loss, indicating capital recovery speed.

$$\mathrm{MDD}_{\mathrm{dur}} = \max(t_{\mathrm{recovery}} - t_{\mathrm{peak}}) \tag{19}$$

Ulcer index combines the depth and duration of drawdowns into a single composite metric to evaluate the holding stress of the investment.

$$\mathrm{UI} = \sqrt{\frac{1}{N} \sum_{t=1}^{N} D_t^2} \tag{20}$$

Sharpe ratio measures the excess return generated per unit of total volatility, serving as the industry standard for risk-adjusted performance, where R_f is a 'risk-free' asset.

$$\mathrm{SR} = \frac{\mu - R_f}{\sigma} \tag{21}$$

Sortino ratio modifies the Sharpe ratio to penalise only downside volatility, making it ideal for investors who view only negative fluctuations as true risk.

$$\mathrm{Sortino} = \frac{\mu - R_f}{\sigma_d} \tag{22}$$

Calmar ratio evaluates the annualised return relative to the meaxmium drawdown, highlighting the specific tradeoff between reward and severe equity drops.

$$\mathrm{Calmar} = \frac{\mathrm{AR}}{\mathrm{MDD}} \tag{23}$$

Profit factor shows the total money made for every dollar lost, acting as a pure, direct measure of absolute trading profitability.

$$\mathrm{PF} = \frac{\sum \text{Gross Profits}}{\sum |\text{Gross Losses}|} \tag{24}$$

Payoff factor compares the absolute size of the average winning trade to the average losing trade, indicating the system's risk-to-reward efficiency.

$$\mathrm{Payoff} = \frac{\text{Average Win}}{|\text{Average Loss}|} \tag{25}$$

Win rate calculates the percentage of trades that result in a profit, reflecting the core accuracy of the strategy's entry and exit signals.

$$W = \frac{N_{\mathrm{wins}}}{N_{\mathrm{trades}}} \tag{26}$$

Market exposure measures the fraction of time capital is actively deployed in the market as opposed to sitting safely in cash.

$$\mathrm{Exposure} = \frac{T_{\mathrm{invested}}}{T_{\mathrm{total}}} \tag{27}$$

Average win calculates the typical gain realised on a successful transaction.

$$\text{AW} = \frac{1}{N_{\text{wins}}} \sum_{i=1}^{N_{\text{wins}}} \text{Profit}_i \tag{28}$$

Average loss calculates the typical capital deficit incurred on an unsuccessful transaction.

$$\text{AL} = \frac{1}{N_{\text{losses}}} \sum_{i=1}^{N_{\text{losses}}} \text{Loss}_i \tag{29}$$

2.5 Data

The empirical analysis is based on high-frequency foreign exchange data downloaded from Dukascopy. For each currency pair, I collect separate bid and ask quote data for both legs of the pair. The raw files are stored as monthly parquet files and loaded into Python through a reproducible data pipeline. The sample covers the period from January 2024 to December 2025, giving two full calendar years of tick-level FX observations.

The three currency-pair constructions used in the project are

AUD/USD–NZD/USD, EUR/NOK–EUR/SEK, GBP/USD–EUR/USD.

These pairs were selected because they represent economically related exchange rates while differing substantially in liquidity, transaction costs, and empirical dependence. AUD/USD and NZD/USD are both Asia-Pacific currencies quoted against the US dollar. EUR/NOK and EUR/SEK connect two Nordic currencies against the euro, but are less liquid and have materially wider spreads. GBP/USD and EUR/USD are highly liquid major currency pairs sharing the US dollar leg.

Using bid and ask data rather than only mid-prices is important in a high-frequency setting. At low frequencies, the spread may be treated as a small implementation detail, but at the tick level it is central to the economics of the strategy. A model may identify statistically significant mean reversion and still fail economically if the expected move is smaller than the cost of trading. For this reason, the data set is constructed to preserve bid prices, ask prices, mid-prices, and spread-based transaction-cost estimates throughout the pipeline.

2.6 Pre-processing

The raw Dukascopy quote data are not used directly in the statistical models. Instead, the data are transformed into synchronized pair-level observations through a custom Python pipeline. This preprocessing is important because high-frequency financial data are irregularly spaced, asynchronous, and strongly affected by bid–ask spreads and microstructure noise. The main preprocessing steps are described below.

2.6.1 Bid–Ask and Mid prices

For each currency leg, bid and ask quote streams are first sorted by timestamp. Since bid and ask updates do not necessarily arrive at exactly the same time, they are aligned using a backward as-of merge. This means that, at each timestamp, the most recent available quote is used, avoiding any use of future information.

The mid-price is then computed as

$$P_t = \frac{P_t^{\text{ask}}}{P_t^{\text{bid}}} \tag{30}$$

which produces a single representative price series for each asset while still preserving the original bid and ask quotes for transaction-cost calculations.

2.6.2 Spread

The bid–ask spread is central in HFT because expected profits are often small relative to trading costs. For each asset, the half-spread is measured in basis points as

$$\text{HS}_t = \frac{1}{2}\frac{P_t^{\text{ask}}}{P_t^{\text{bid}}} \times 10000 \tag{31}$$

The half-spread gives an estimate of the cost of crossing the market for one leg of the trade. Since pairs trading requires trading both legs, the combined spread cost is an important part of the later backtest.

2.7 Pairs trading

We choose to financial instruments, as AUDUSD and NZDUSD in this paper. Let us define the spread z_t as

$$z_t = p_t^{(1)} - \beta p_t^{(2)} \tag{32}$$

where β can be estimated by the Engle-Granger method, which means using an Ordinary Least Squares (OLS) regression followed by an Augmented Dickey-Fuller (ADF) test on the residuals to verify that z_t is stationary. It is also possible to esimate this directly in the HMM model, but easier to do so globally at first.

Thereafter we can fit an AR- or ARCH-HMM on the spread-process

$$z_t = \mu^k + \rho^k(z_{t-1} - \mu^k) + \varepsilon_t^k \tag{33}$$

where ρ^k measures how fast we expect the mean reversion to be in regime k. Values of ρ^k close to 0 indicate a faster mean reversion, while values closer to 1 indicate a slower mean reversion. Negative values of ρ^k would indicate even stronger mean reversion but possibly less realistic empirically.

The trading rule in pairs trading can be formulated as follows. In a drifting regime we do nothing. In a mean-reverting regime, say for regime $k = 2$ we trade by.

$$\begin{cases} z_t > \mu^2 + c\sigma^2 \Rightarrow \text{short AUDUSD and long NZDUSD,} \\ z_t < \mu^2 - c\sigma^2 \Rightarrow \text{long AUDUSD and short NZDUSD,} \\ \qquad z_t \approx \mu^2 \Rightarrow \text{close the position.} \end{cases} \tag{34}$$

While for one unit of AUDUSD we trade $\hat{\beta}$ units of NZDUSD, such that the position in main is exposed to the relative movement between AUDUSD and NZDUUSD, and not toward the general level in the currency market.

For a soft HMM version the positions can be scaled with the probabilites of the model being in a certain mean-reverting regime

$$w(s_t) = \mathbb{P}(s_t \text{is in a mean-reverting regime} \mid \mathcal{F}_\sqcup) \tag{35}$$

It may also be natural to scale the positions with esitmated volatility.

2.8 Back-testing

Back-testing is used to evaluate whether the model signals translate into economically meaningful trading performance. This is important because a model can fit historical returns well without producing a tradable strategy. In high-frequency trading, the expected profit per trade is often small, so even modest execution costs can eliminate an apparent statistical edge.

For this reason, the backtest includes realistic market frictions. The most important is the bid–ask spread: a trader generally buys at the ask and sells at the bid, rather than trading for free at the mid-price. Commissions and other execution costs are also included, since frequent position changes can make small costs accumulate quickly. Financing and margin costs are relevant when the strategy holds short or leveraged positions.

The strategies are evaluated in a walk-forward structure, where parameters are estimated on past data and then applied to later out-of-sample periods. This avoids look-ahead bias and gives a more realistic test of whether the baseline, AR, and Markov-switching strategies would have worked under implementable trading conditions.

3 Results

3.1 Training the HMM

The Hidden Markov Model is trained on the hedged spread-return series constructed from each currency pair. The regimes are sorted by estimated variance, so that the low-variance state is interpreted as a relatively safe trading environment and the high-variance state as a dangerous or stressed environment. This gives the model a direct economic interpretation: it is not primarily used to forecast the direction of the exchange rate, but to decide whether the current market state is suitable for applying a mean-reversion trading rule.

Across the experiments, the Markov model generally identifies periods of elevated spread-return variance and reduces exposure during these episodes. The key empirical question is therefore whether this regime filter improves trading performance, or whether it merely removes both downside and upside. The results below show that the HMM is useful as a defensive exposure filter, but does not consistently outperform simpler strategies.

3.2 Data and pair diagnostics

Table 1 summarizes the constructed volume-bar samples and liquidity conditions across the three tested currency pairs. The pairs differ substantially in trading costs. GBPUSD/EURUSD is by far the cheapest pair to trade, with a median round-trip cost of only 0.46 bps. AUDUSD/NZDUSD is still relatively liquid, while EURNOK/EURSEK is materially more expensive, with a median round-trip cost of 3.89 bps. This difference becomes important in the backtest, since high-frequency mean-reversion strategies are highly sensitive to spread costs.

Table 1: Sample and liquidity diagnostics across currency pairs. The sample consists of volume bars constructed from Dukascopy bid–ask data over 2024–2025. Spreads are reported in basis points. Note that AUDNZD is short for AUDUSD/NZD, NOKSEK is short for EURNOK/EURSEK, and GBPEUR is short for GBPUSD/EURUSD.

Metric	**AUDNZD**	**NOKSEK**	**GBPEUR**
Number of bars	108,093	206,937	114,759
Calendar span	729 days	729 days	729 days
Trading days	523	522	523
Average bars per day	206.7	396.4	219.4
Trading hours covered	0–24 UTC	0–24 UTC	0–24 UTC
Median spread, leg A	0.76	2.07	0.32
Median spread, leg B	0.89	1.78	0.14
P95 spread, leg A	1.04	6.22	0.52
P95 spread, leg B	1.21	5.58	0.35
Median round-trip cost	1.63	3.89	0.46
Tightest 4-hour window	11:00–14:00	09:00–12:00	11:00–14:00
Widest 4-hour window	20:00–23:00	20:00–23:00	20:00–23:00

Table 2 reports the main return-distribution and pair-dependence diagnostics. All six currency return series exhibit strong non-Gaussian behaviour, with high kurtosis and significant Ljung–Box tests on squared returns. This confirms the presence of volatility clustering and motivates regime-aware modelling. GBPUSD/EURUSD has the strongest dependence, with a Pearson correlation of 0.5230 and a median rolling correlation of 0.535. EURNOK/EURSEK has the weakest dependence, despite having the largest number of bars.

Table 2: Return and pair-dependence diagnostics. Returns are measured in basis points. The Ljung–Box test is applied to squared returns at lag 20. Note that AUDNZD is short for AUDUSD/NZD, NOKSEK is short for EURNOK/EURSEK, and GBPEUR is short for GBPUSD/EURUSD.

	AUDNZD		**NOKSEK**		**GBPEUR**	
Metric	**AUDUSD**	**NZDUSD**	**EURNOK**	**EURSEK**	**GBPUSD**	**EURUSD**
Mean return	-0.002	-0.009	0.003	-0.001	0.005	0.005
Volatility	4.106	4.076	2.606	2.270	2.990	3.085
Skewness	-0.176	-0.342	0.178	-0.128	-0.325	-0.269
Kurtosis	17.533	22.076	16.922	21.342	14.698	28.557
5th percentile	-6.346	-6.569	-4.062	-3.424	-4.672	-4.744
95th percentile	6.249	6.547	4.072	3.441	4.666	4.687
Minimum	-115.391	-122.287	-50.892	-65.402	-75.882	-107.932
Maximum	69.711	76.045	70.470	44.625	46.876	63.748
Ljung–Box p on r_t^2	0.000	0.000	0.000	0.000	0.000	0.000
Pair relationship						
Pearson correlation		0.3720		0.2038		0.5230
Spearman correlation		0.3308		0.1576		0.4940
OLS hedge ratio β		0.3693		0.1776		0.5395
Rolling corr, min.		-0.041		-0.221		0.086
Rolling corr, median		0.357		0.194		0.535
Rolling corr, max.		0.768		0.521		0.810

3.3 Cointegration and walk-forward optimization

Table 3 combines the main cointegration and walk-forward optimization results. The full-sample cointegration tests are not sufficient on their own. AUDUSD/NZDUSD and

GBPUSD/EURUSD do not reject the no-cointegration null at conventional levels, but both show intermittent local mean reversion in rolling windows. EURNOK/EURSEK rejects the full-sample null, but its estimated half-life is extremely long, making the relationship economically weak for high-frequency trading.

The walk-forward results reinforce this distinction. AUDUSD/NZDUSD produces the strongest out-of-sample profile, with 17 positive windows out of 21 and an average OOS return of 2.65%. GBPUSD/EURUSD is profitable but weaker, with 11 positive windows and a mean OOS return of 0.45%. EURNOK/EURSEK fails completely, producing no positive OOS windows and an average OOS return of -4.19%.

Table 3: Cointegration and walk-forward optimization summary across currency pairs. Rolling cointegration uses 2,000-bar windows stepped every 200 bars. Note that AUDNZD is short for AUDUSD/NZD, NOKSEK is short for EURNOK/EURSEK, and GBPEUR is short for GBPUSD/EURUSD.

Metric	**AUDNZD**	**NOKSEK**	**GBPEUR**
Full-sample cointegration			
Cointegration p-value	0.8657	0.0140	0.1118
Half-life	2415.8 bars	4597.2 bars	2472.3 bars
Hedge ratio β	0.6641	0.0714	0.6657
Rolling cointegration			
Number of rolling windows	531	1025	564
Share with $p < 0.05$	9.8%	13.1%	8.2%
Share with $p < 0.10$	16.8%	21.1%	15.6%
Median half-life	32.5 bars	71.8 bars	72.2 bars
Half-life IQR	19–65	44–125	44–129
Mean rolling β	0.8363	0.6911	0.8672
Std. dev. rolling β	0.2857	0.6648	0.3686
Walk-forward optimization			
Validation / test design	3m / 1m	3m / 1m	3m / 1m
Trials per window	100	100	150
Number of OOS windows	21	21	21
Mean OOS return per window	2.65%	-4.19%	0.45%
Positive OOS windows	17 / 21	0 / 21	11 / 21
Zero-return OOS windows	2 / 21	4 / 21	5 / 21
Negative OOS windows	2 / 21	17 / 21	5 / 21
Best OOS window	12.30%	0.00%	3.06%
Worst OOS window	-0.66%	-23.74%	-1.50%
Typical entry threshold	$z \approx 1.4$	$z \approx 2.5$	$z \approx 1.1$–1.5
Typical exit threshold	$z \approx 0.2$–0.5	$z \approx -0.5$–0.0	$z \approx 0.0$–0.5
Typical danger threshold	0.95	0.50–0.65	0.85–0.95
Typical AR limit	0.96–0.97	0.98–0.99	0.97–0.99

3.4 Trading performance

Table 4 reports the core performance metrics for the baseline, AR-filtered, and Markov-switching AR strategies. The main result is clear: the HMM does not consistently improve performance relative to simpler rules. Instead, it primarily reduces market exposure. This is useful when the pair is losing money, but costly when the baseline strategy is profitable.

For AUDUSD/NZDUSD, all strategies are profitable, but the baseline dominates. It earns 9286.14 bps with a Sharpe ratio of 6.88 and a Calmar ratio of 21.40. The AR and MS-AR filters reduce volatility and exposure, but also reduce total return. The Markov-switching model therefore acts as a conservative filter rather than a return enhancer.

EURNOK/EURSEK is not economically viable under the tested rules. All three strategies lose money. The MS-AR model reduces the loss from -14035.70 bps to -8796.04 bps and lowers exposure from 42.70% to 25.11%, but it does not convert the pair into a profitable strategy. This confirms that statistical structure is not enough when transaction costs and weak dependence dominate the signal.

GBPUSD/EURUSD is the most liquid and most correlated pair in the sample. It produces positive returns across all three model specifications, but again the simpler models perform best. The baseline achieves the highest total return and best Calmar ratio, while the AR model preserves a similar Sharpe ratio with lower exposure. The MS-AR model reduces exposure further, but total return falls to 932.85 bps and the Sharpe ratio declines to 1.24.

Table 4: Core strategy performance across currency pairs. Performance is reported net of the backtesting assumptions used in the simulation. Returns, volatility, and drawdowns are reported in basis points.

	AUDUSD/NZDUSD			**EURNOK/EURSEK**			**GBPUSD/EURUSD**		
Metric	**Base.**	**AR**	**MS-AR**	**Base.**	**AR**	**MS-AR**	**Base.**	**AR**	**MS-AR**
Total return	9286.14	7695.77	5567.68	-14035.70	-10581.39	-8796.04	2059.43	1677.66	932.85
Annual return	5643.53	4677.00	3383.69	-8544.23	-6441.42	-5354.59	1251.59	1019.58	566.93
Positive months	100.00%	90.00%	80.00%	0.00%	0.00%	0.00%	70.00%	55.00%	45.00%
Annual volatility	819.72	724.16	686.87	803.51	632.82	550.25	572.05	478.18	458.24
Max drawdown	-263.67	-263.67	-281.26	-14096.99	-10613.91	-8838.14	-372.00	-569.59	-629.09
Ulcer index	59.56	74.47	79.16	8439.75	6493.59	5343.53	129.96	185.14	232.89
Sharpe ratio	6.88	6.46	4.93	-10.63	-10.18	-9.73	2.19	2.13	1.24
Sortino ratio	8.59	7.05	5.23	-8.10	-6.51	-5.52	2.51	1.99	1.12
Calmar ratio	21.40	17.74	12.03	0.61	0.61	0.61	3.36	1.79	0.90
Profit factor	1.11	1.12	1.09	0.85	0.83	0.82	1.03	1.04	1.02
Number of trades	3158	2452	2254	1702	1233	938	2070	1484	1360
Win rate	49.55%	49.62%	49.44%	48.82%	48.82%	48.68%	49.39%	49.41%	49.22%
Market exposure	60.97%	46.69%	45.01%	42.70%	31.38%	25.11%	62.59%	41.44%	39.17%
Skewness	1.17	1.48	0.76	-1.63	-2.58	-3.62	-0.05	-0.02	-0.19
Kurtosis	34.41	51.53	21.08	52.00	82.88	141.09	19.01	29.03	29.34

3.5 Summary of results

The empirical results show that Hidden Markov regime filtering is more effective at reducing exposure than at improving returns. In all three pairs, the MS-AR strategy trades less than the baseline and usually produces lower volatility. However, this defensive behaviour does not translate into superior risk-adjusted performance.

The strongest pair is AUDUSD/NZDUSD, where the baseline strategy performs best and the Markov model removes too much profitable exposure. GBPUSD/EURUSD is also profitable, but the MS-AR filter again underperforms the simpler alternatives. EURNOK/EURSEK fails economically across all specifications; here the HMM is useful only in the limited sense that it reduces the size of the loss.

Overall, the results suggest that the HMM identifies high-risk market states, but the identified regimes are not sufficiently aligned with profitable trading opportunities. The model is therefore valuable as a diagnostic and defensive tool, but not as a standalone source of superior high-frequency pairs-trading performance.

4 Conclusion

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5 Appendix

5.1 Data pipeline

Table 5: Chronological overview of the data processing and modelling pipeline.

Step	Stage	Description
1	Download data	Download monthly Dukascopy tick data for each currency leg.
2	Separate quote sides	Store bid and ask files separately for each asset and month.
3	Load files	Load the relevant parquet files into Python.
4	Filter observations	Restrict the data to the chosen trading days and active trading hours.
5	Sort timestamps	Sort all bid and ask quote streams chronologically.
6	Align bid–ask quotes	Merge bid and ask streams within each asset using backward as-of alignment.
7	Compute mid-prices	Compute mid-prices from bid and ask quotes.
8	Compute half-spreads	Compute half-spreads in basis points for each asset.
9	Aggregate bars	Aggregate raw ticks into event-based volume bars.
10	Preserve continuity	Carry cumulative volume across monthly file boundaries.
11	Synchronize pair legs	Merge the two assets into synchronized pair-level observations.
12	Compute log-prices	Transform mid-prices into log-prices.
13	Compute returns	Compute one-bar log returns for both assets.
14	Estimate hedge ratio	Estimate static and rolling hedge ratios between the two log-price series.
15	Construct spread	Construct the log spread using the estimated hedge ratio.
16	Standardize spread	Compute rolling spread mean, rolling spread volatility, and the spread z-score.
17	Compute spread returns	Compute hedged spread returns using lagged hedge ratios.
18	Run diagnostics	Produce descriptive, liquidity, correlation, and cointegration diagnostics.
19	Fit AR model	Estimate the autoregressive benchmark model.
20	Fit Markov model	Estimate hidden Markov regime probabilities using spread returns.
21	Walk-forward prediction	Generate out-of-sample signals and regime probabilities.
22	Optimize rules	Use walk-forward optimization to select trading thresholds.
23	Backtest strategies	Evaluate baseline, AR-filtered, and MS-AR strategies net of costs.
24	Summarize performance	Produce tearsheets, figures, and strategy comparison tables.

5.2 Exploratory data analysis

5.2.1 AUDUSD/NZDUSD

Figure 1: Exploratory data analysis for AUDUSD/NZDUSD. The figure summarizes intraday spreads, return volatility, mean returns, activity patterns, weekday effects, rolling realized volatility, rolling correlation, return distributions, normality diagnostics, cross-pair return scatter, autocorrelation, rolling volatility distribution, and the relationship between spread and volatility.

5.2.2 EURNOK/EURSEK

Figure 2: Exploratory data analysis for EURNOK/EURSEK. The figure summarizes intraday spreads, volatility, mean returns, bar counts, weekday effects, rolling realized volatility, rolling correlation, distributional features, normality diagnostics, cross-pair return scatter, autocorrelation, rolling volatility, and the spread-volatility relationship.

5.2.3 GBPUSD/EURUSD

Figure 3: Exploratory data analysis for GBPUSD/EURUSD. The figure summarizes intraday liquidity and volatility patterns, hourly and weekday activity, rolling realized volatility, rolling correlation, return distributions, normality diagnostics, cross-pair return dependence, autocorrelation, and the relationship between spread and volatility.

5.3 Spread diagnostics

5.3.1 AUDUSD/NZDUSD

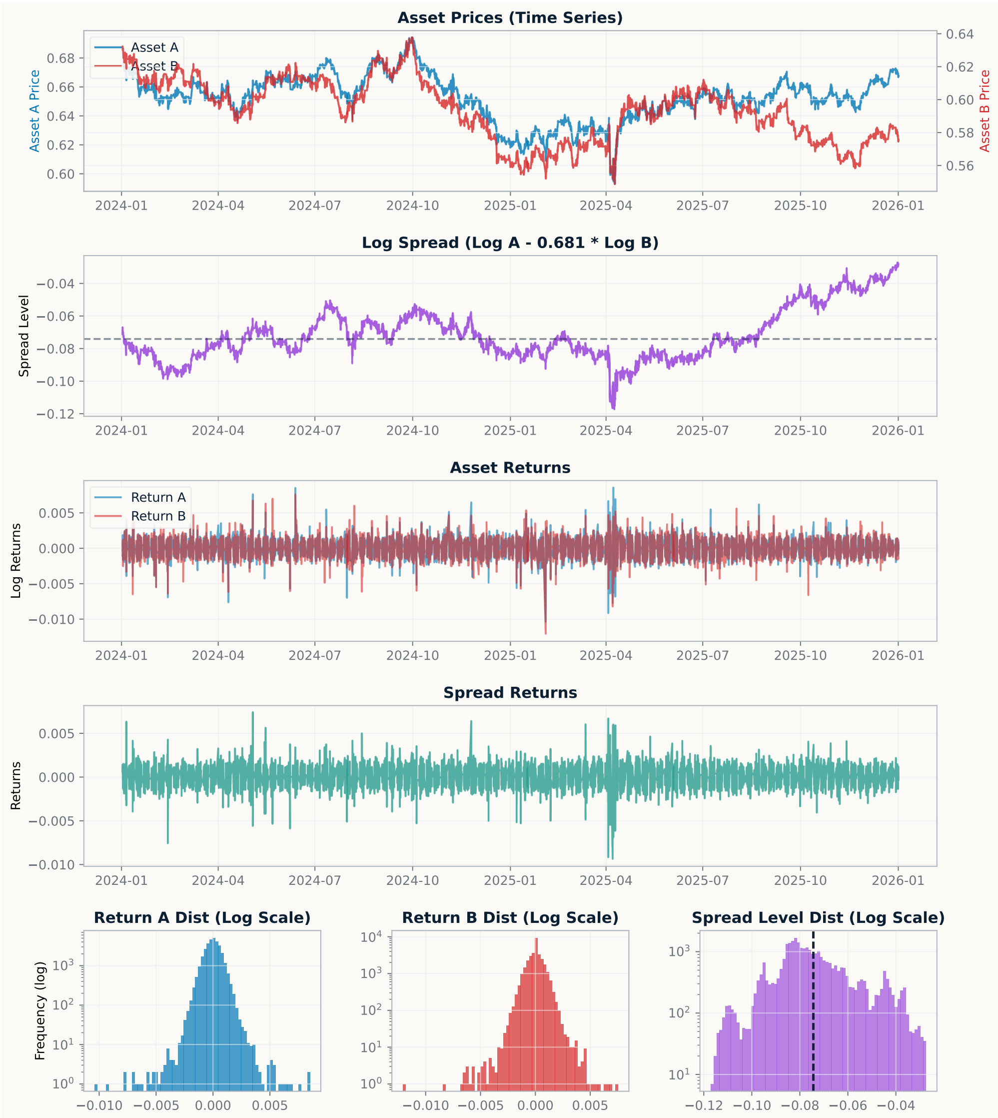

Figure 4: Spread diagnostics for AUDUSD/NZDUSD. The figure shows the two asset price series, the estimated log spread, asset returns, spread returns, and distributional diagnostics for both returns and spread levels, providing a visual check of the pairs-trading signal construction.

5.3.2 EURNOK/EURSEK

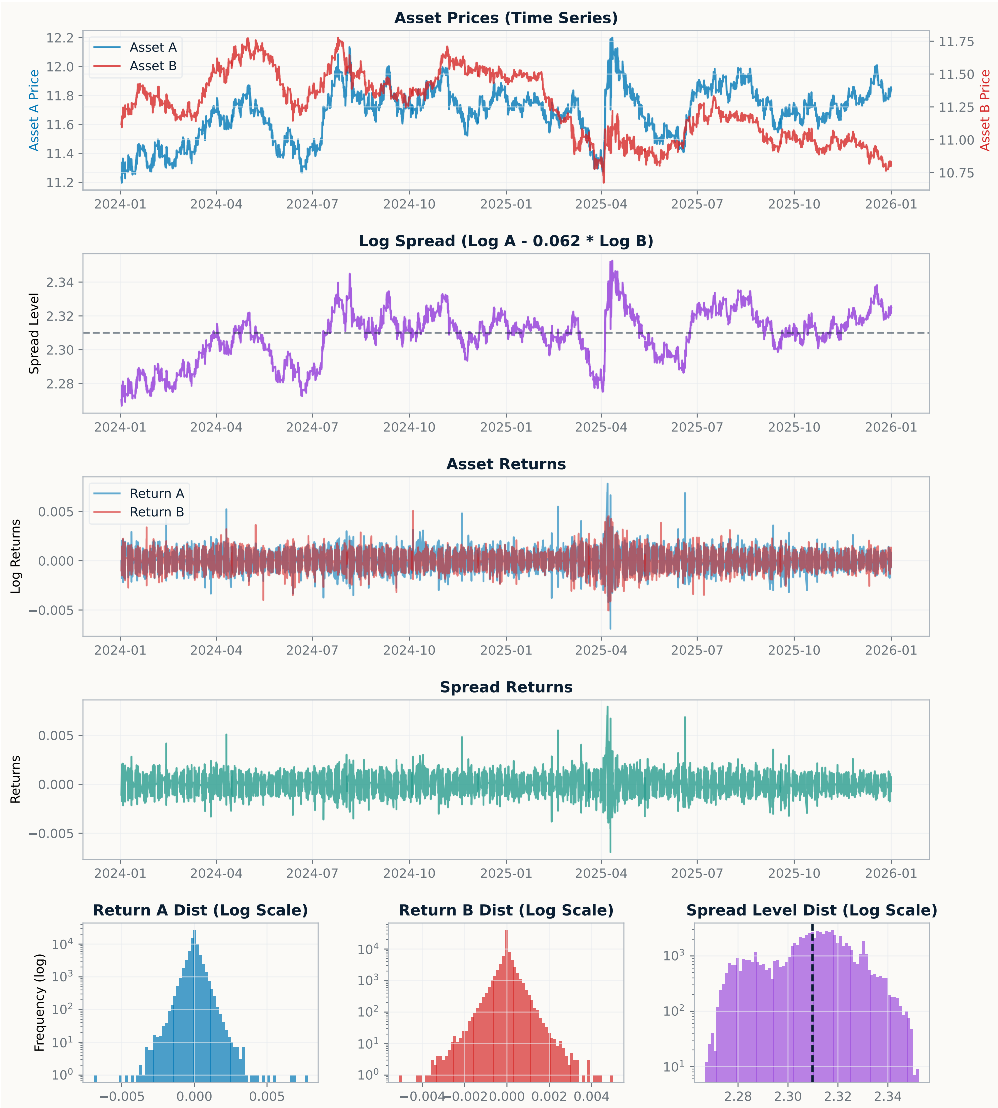

Figure 5: Spread diagnostics for EURNOK/EURSEK. The figure presents the asset price paths, fitted log spread, asset returns, spread returns, and distributional diagnostics, allowing visual inspection of spread stability, return asymmetry, and tail behaviour.

5.3.3 GBPUSD/EURUSD

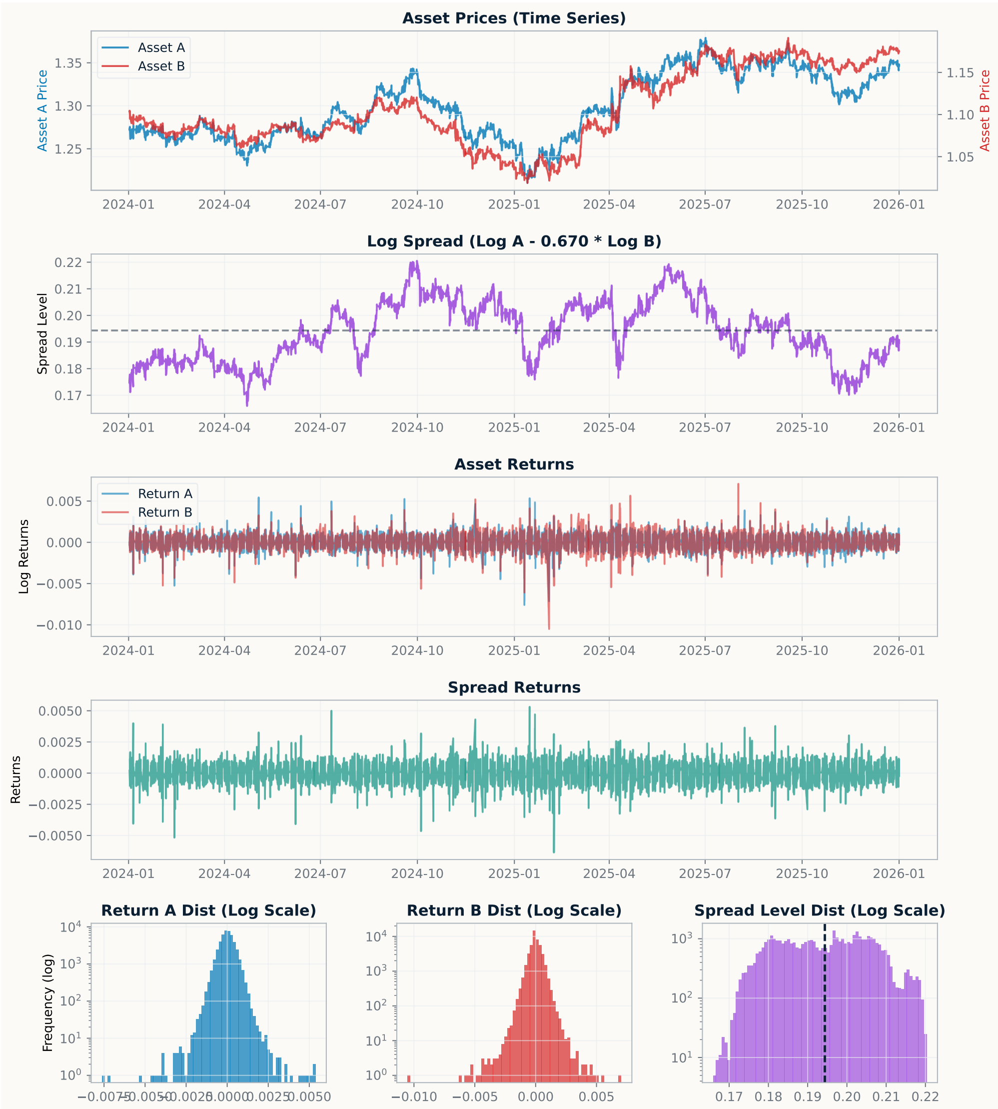

Figure 6: Spread diagnostics for GBPUSD/EURUSD. The figure shows the two asset price series, estimated log spread, asset returns, spread returns, and distributional diagnostics, providing a visual basis for assessing the stability and tradability of the constructed spread.

5.4 Backtest performance

5.4.1 AUDUSD/NZDUSD

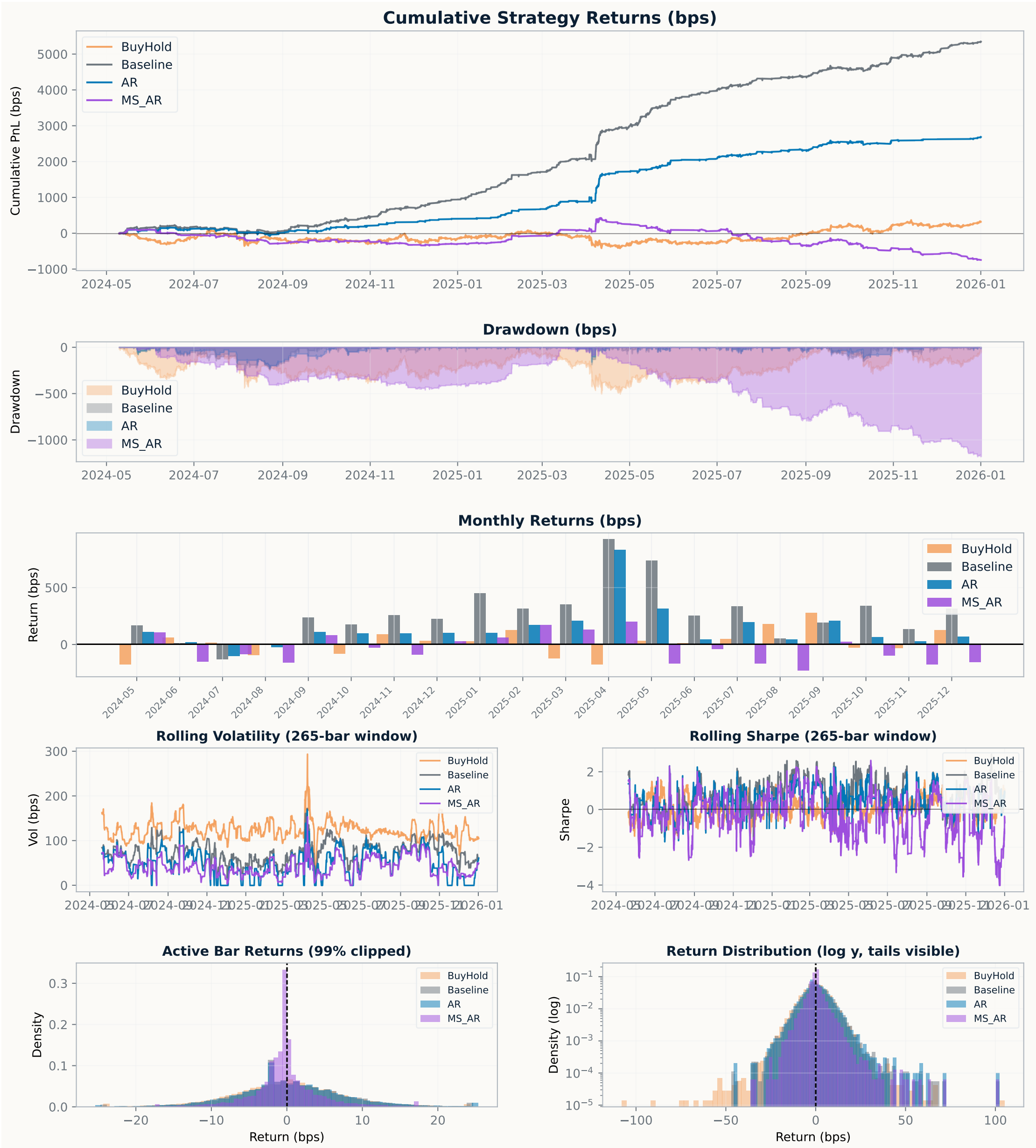

Figure 7: Backtest performance for AUDUSD/NZDUSD across the baseline, AR, and Markov-switching AR strategies. The figure compares cumulative strategy returns, drawdowns, monthly returns, rolling volatility, rolling Sharpe ratios, active-bar return distributions, and tail behaviour on a log-density scale.

5.4.2 EURNOK/EURSEK

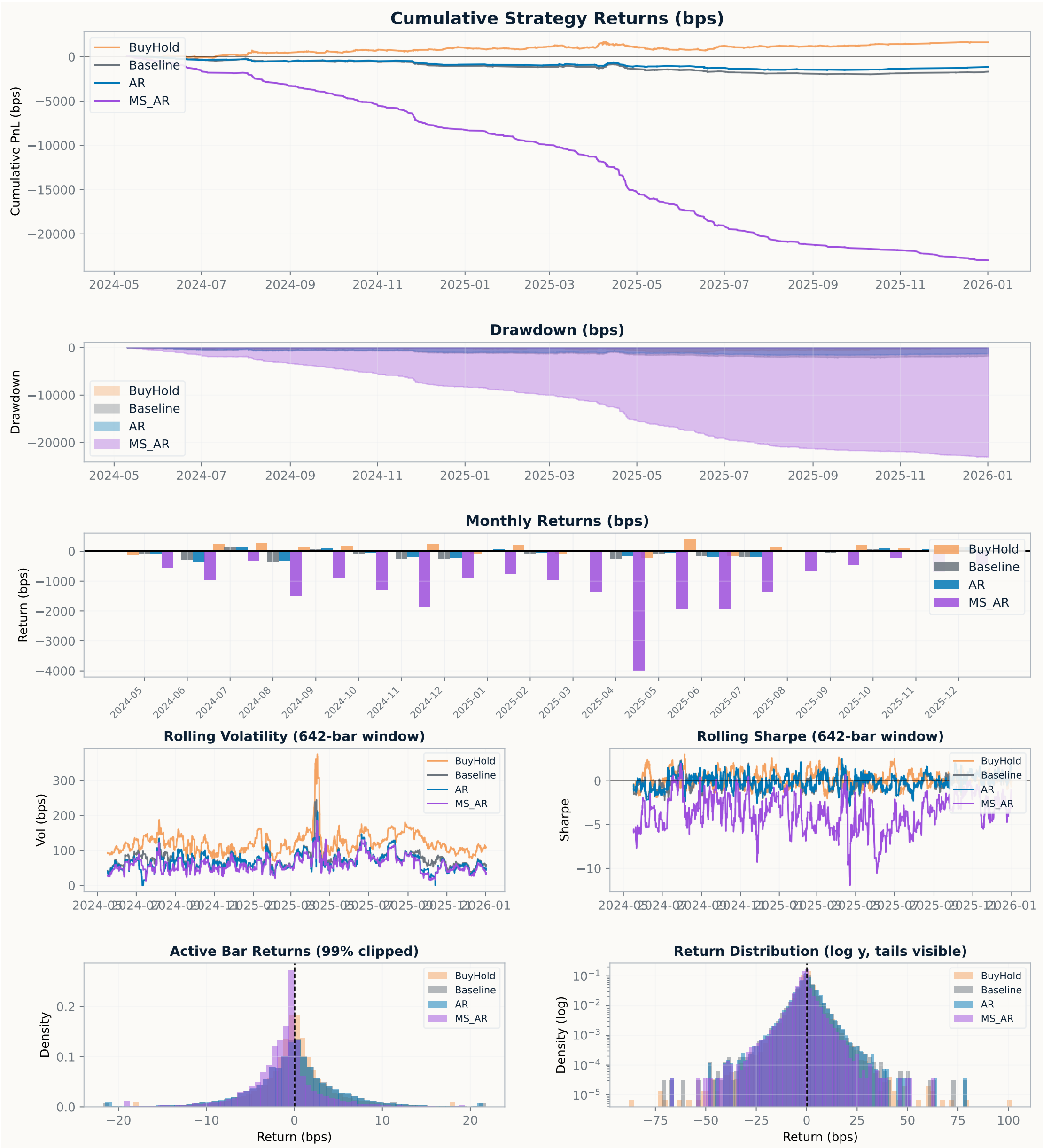

Figure 8: Backtest performance for EURNOK/EURSEK across the baseline, AR, and Markov-switching AR strategies. The figure compares cumulative returns, drawdowns, monthly return contributions, rolling volatility, rolling Sharpe ratios, active-bar return densities, and tail behaviour.

5.4.3 GBPUSD/EURUSD

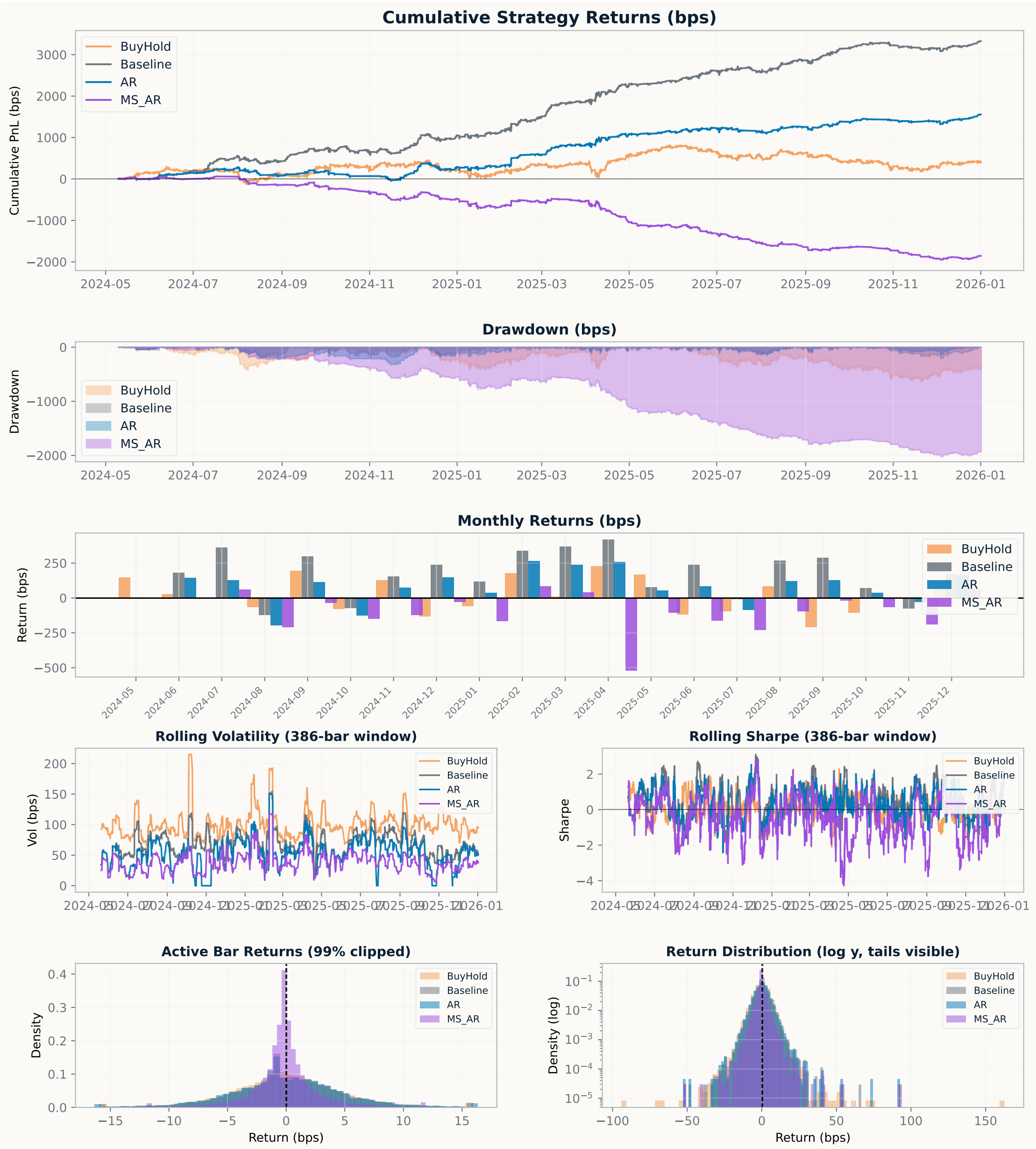

Figure 9: Backtest performance for GBPUSD/EURUSD across the baseline, AR, and Markov-switching AR strategies. The figure compares cumulative returns, drawdowns, monthly returns, rolling volatility, rolling Sharpe ratios, active-bar return densities, and tail behaviour under the competing model specifications.

5.5 Currency pair tables

Table 6: Full strategy tearsheets across all currency pairs. Returns, volatility, drawdowns, VaR, CVaR, average wins, and losses are reported in basis points.

	AUDUSD/NZDUSD			EURNOK/EURSEK			GBPUSD/EURUSD		
Metric	**Base**	**AR**	**MS-AR**	**Base**	**AR**	**MS-AR**	**Base**	**AR**	**MS-AR**
Total return	9286.14	7695.77	5567.68	-14035.70	-10581.39	-8796.04	2059.43	1677.66	932.85
Annual return	5643.53	4677.00	3383.69	-8544.23	-6441.42	-5354.59	1251.59	1019.58	566.93
Best month	1971.98	1971.98	1297.72	-79.73	0.00	0.00	431.52	410.25	311.83
Worst month	62.56	-73.87	-70.33	-3224.25	-2611.60	-2504.00	-277.46	-130.81	-140.36
Positive months	100.00%	90.00%	80.00%	0.00%	0.00%	0.00%	70.00%	55.00%	45.00%
Annual volatility	819.72	724.16	686.87	803.51	632.82	550.25	572.05	478.18	458.24
Max drawdown	-263.67	-263.67	-281.26	-14096.99	-10613.91	-8838.14	-372.00	-569.59	-629.09
Max drawdown duration	5525	11954	12623	177619	177619	177666	40516	47494	56854
Ulcer index	59.56	74.47	79.16	8439.75	6493.59	5343.53	129.96	185.14	232.89
VaR 95%	-5.23	-4.48	-4.39	-3.38	-2.36	-1.76	-3.66	-3.00	-2.88
CVaR 95%	-7.99	-7.28	-7.11	-6.62	-5.24	-4.50	-5.65	-4.98	-4.83
Sharpe ratio	6.88	6.46	4.93	-10.63	-10.18	-9.73	2.19	2.13	1.24
Sortino ratio	8.59	7.05	5.23	-8.10	-6.51	-5.52	2.51	1.99	1.12
Calmar ratio	21.40	17.74	12.03	0.61	0.61	0.61	3.36	1.79	0.90
Profit factor	1.11	1.12	1.09	0.85	0.83	0.82	1.03	1.04	1.02
Payoff ratio	1.13	1.14	1.12	0.89	0.87	0.87	1.06	1.06	1.06
Tail ratio	1.14	1.16	1.14	0.86	0.84	0.83	1.05	1.05	1.05
Number of trades	3158	2452	2254	1702	1233	938	2070	1484	1360
Win rate	49.55%	49.62%	49.44%	48.82%	48.82%	48.68%	49.39%	49.41%	49.22%
Market exposure	60.97%	46.69%	45.01%	42.70%	31.38%	25.11%	62.59%	41.44%	39.17%
Average win	3.47	3.49	3.40	2.14	1.94	1.88	2.20	2.27	2.24
Average loss	-3.07	-3.06	-3.04	-2.40	-2.22	-2.16	-2.08	-2.13	-2.12
Skewness	1.17	1.48	0.76	-1.63	-2.58	-3.62	-0.05	-0.02	-0.19
Kurtosis	34.41	51.53	21.08	52.00	82.88	141.09	19.01	29.03	29.34

5.6 Markov Dynamics

5.6.1 AUDUSD/NZDUSD

Figure 10: Estimated Markov-regime dynamics for AUDUSD/NZDUSD. The figure reports rolling estimates of regime-specific variance, expected return, and regime persistence, illustrating how the hidden Markov structure classifies changing market conditions through time.

5.6.2 EURNOK/EURSEK

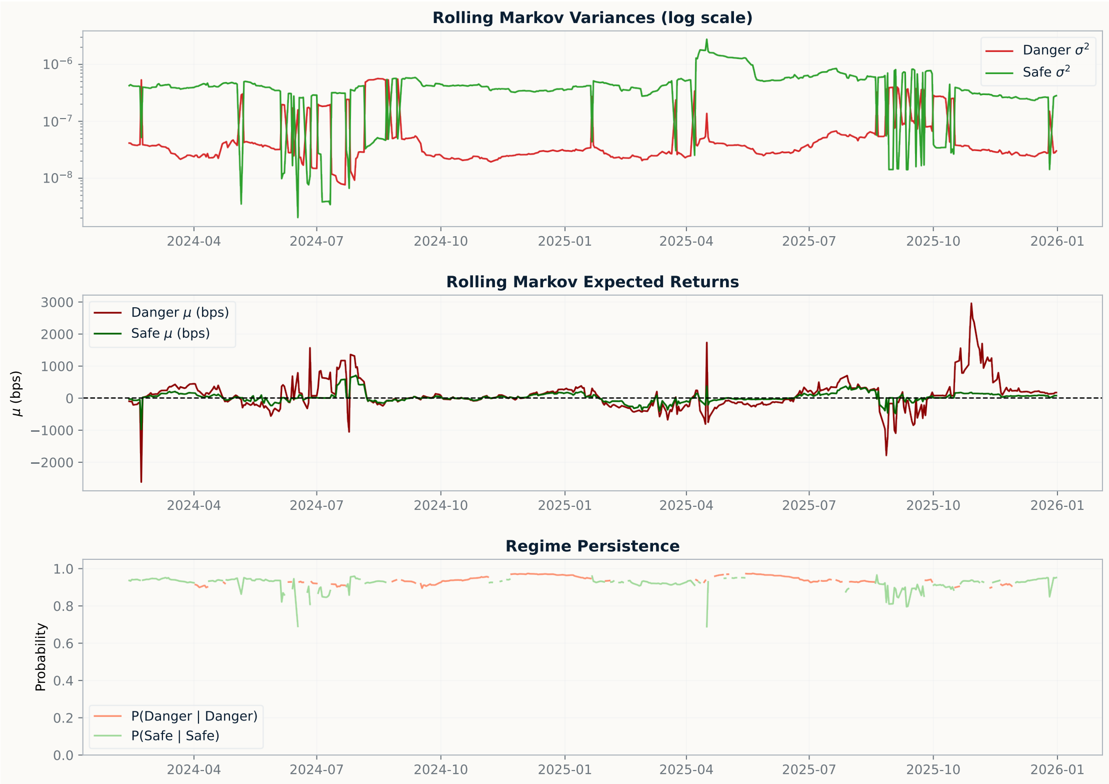

Figure 11: Estimated Markov-regime dynamics for EURNOK/EURSEK. The figure reports rolling estimates of regime-specific variance, expected return, and regime persistence, illustrating how the hidden Markov structure classifies changing market conditions through time.

5.6.3 GBPUSD/EURUSD

Figure 12: Estimated Markov-regime dynamics for GBPUSD/EURUSD. The figure reports rolling estimates of regime-specific variance, expected return, and regime persistence, illustrating how the hidden Markov structure classifies changing market conditions through time.

5.7 Positions and regimes

5.7.1 AUDUSD/NZDUSD

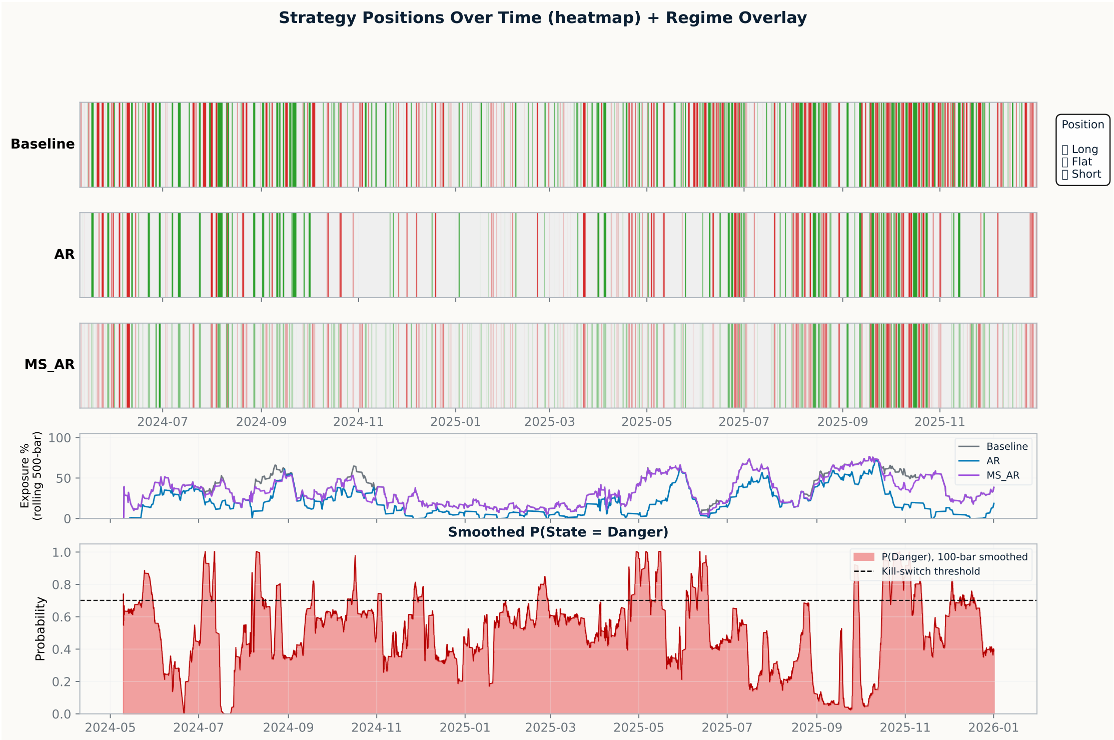

Figure 13: Trading positions and regime overlays for AUDUSD/NZDUSD. The figure shows exposure across the baseline, AR, and Markov-switching AR strategies, together with the smoothed dangerous-regime probability and the resulting long, flat, and short trading states.

5.7.2 EURNOK/EURSEK

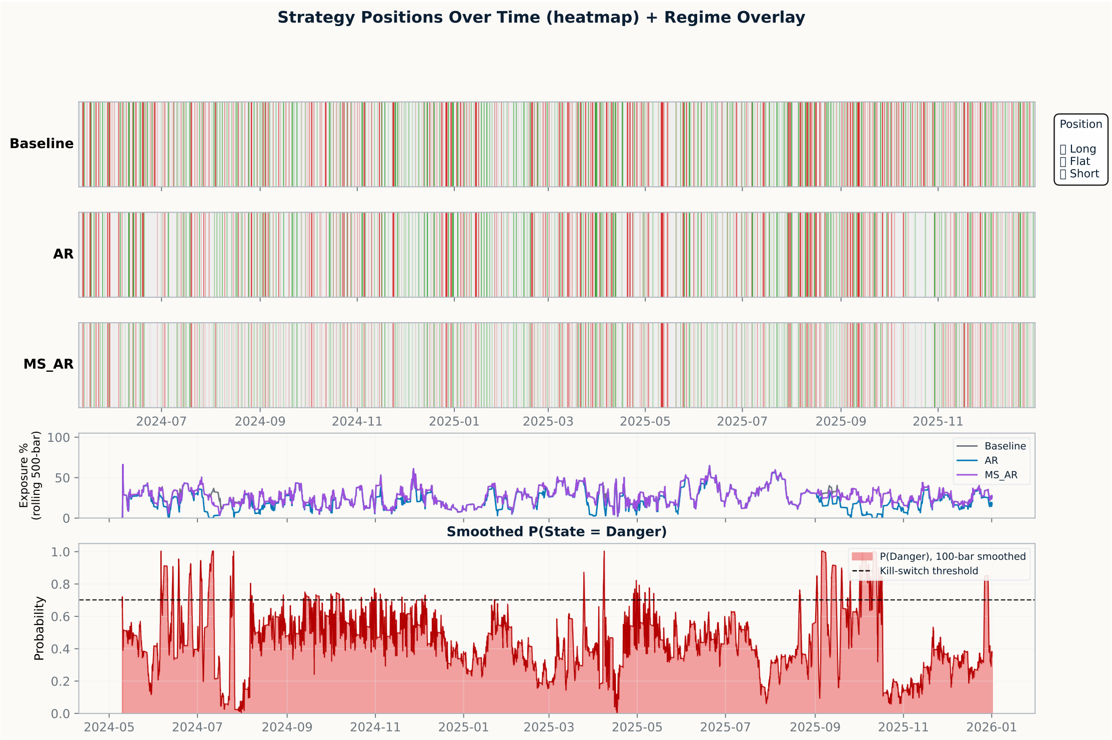

Figure 14: Trading positions and regime overlays for EURNOK/EURSEK. The figure shows exposure across the baseline, AR, and Markov-switching AR strategies, together with the smoothed dangerous-regime probability and the resulting long, flat, and short trading states.

5.7.3 GBPUSD/EURUSD

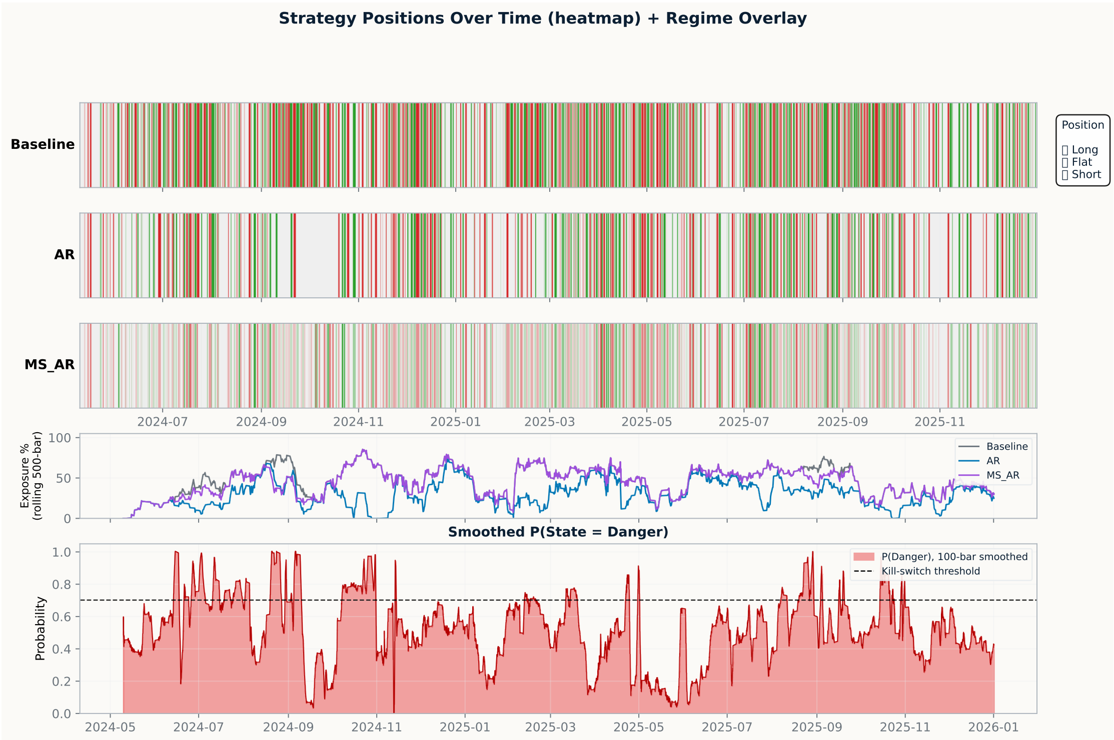

Figure 15: Trading positions and regime overlays for GBPUSD/EURUSD. The figure shows exposure across the baseline, AR, and Markov-switching AR strategies, together with the smoothed dangerous-regime probability and the resulting long, flat, and short trading states.

5.8 Cost impact

5.8.1 AUDUSD/NZDUSD

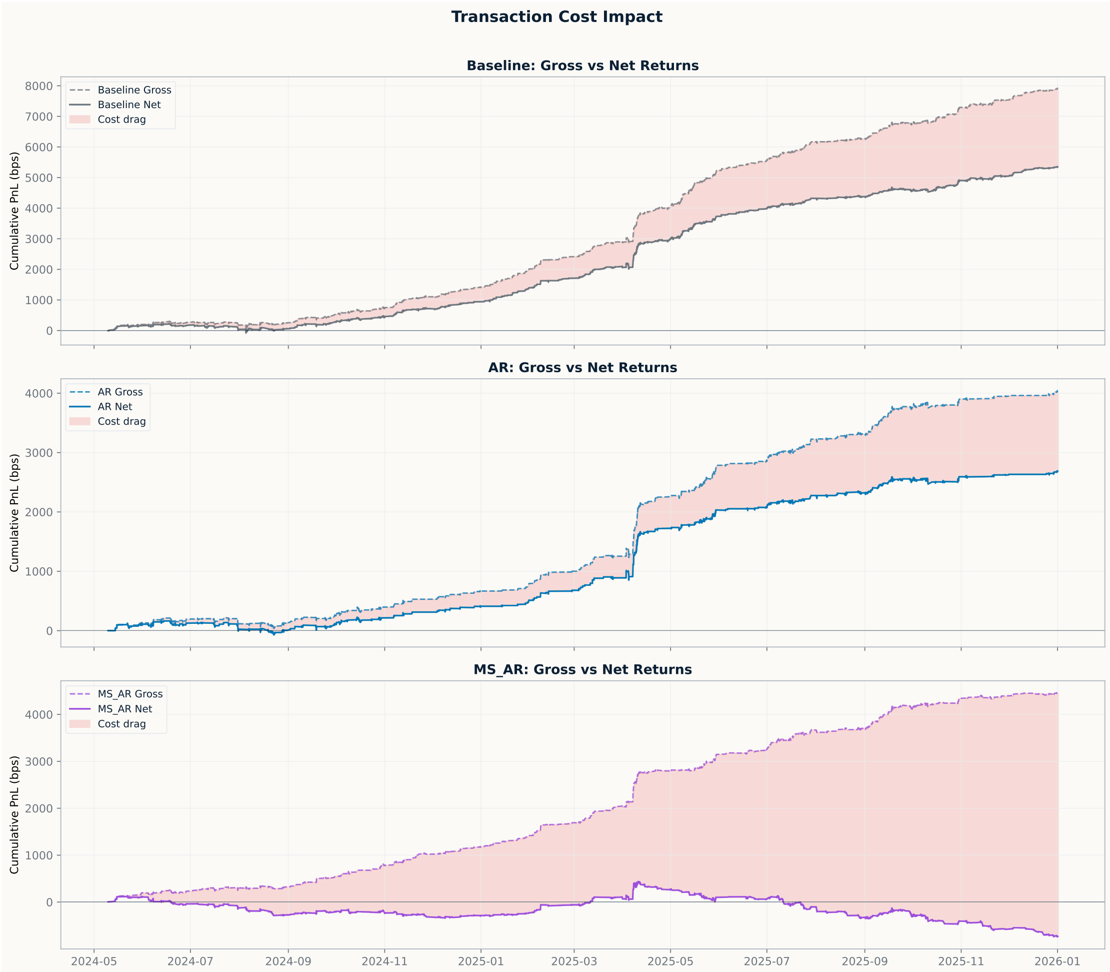

Figure 16: Cost impact for AUDUSD/NZDUSD. The figure shows how transaction costs affect strategy performance for the baseline, AR, and Markov-switching AR strategies, highlighting the sensitivity of cumulative returns to implementation frictions.

5.8.2 EURNOK/EURSEK

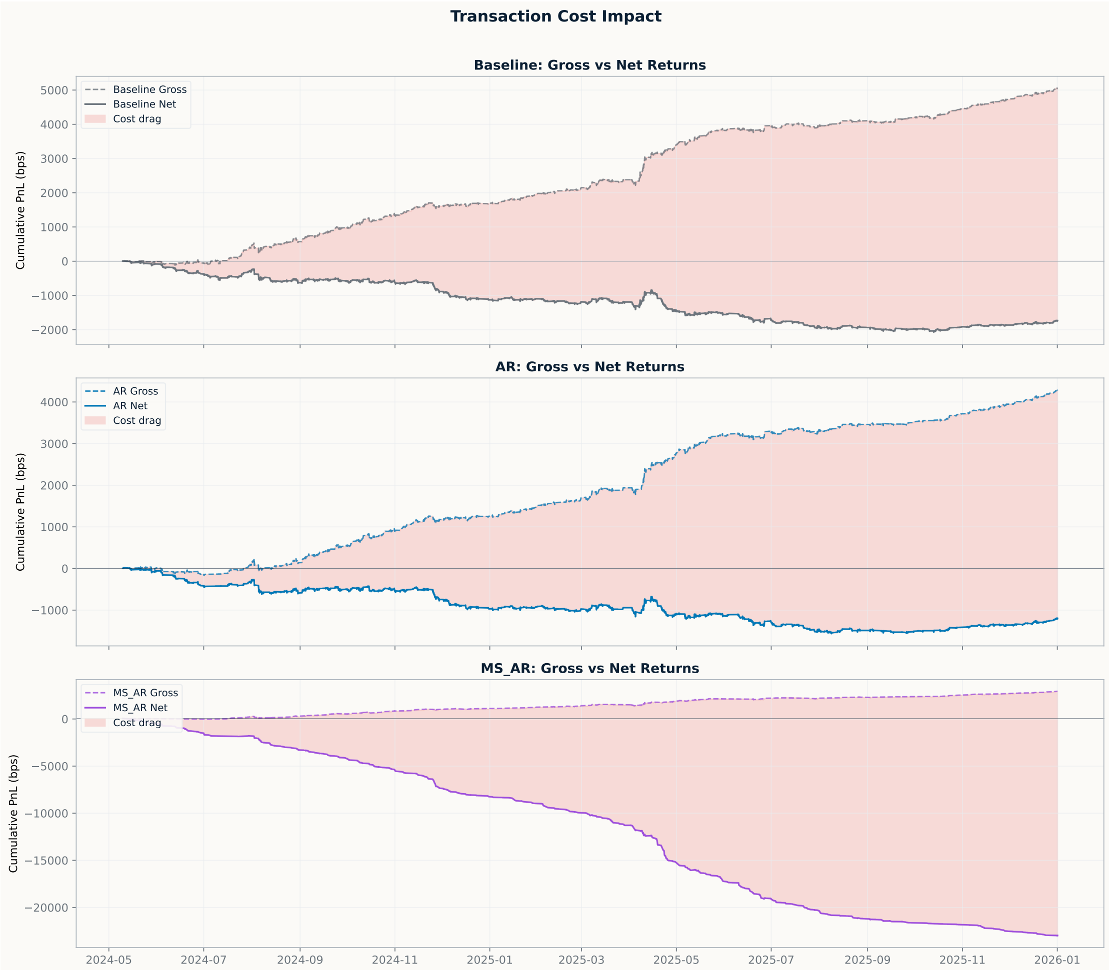

Figure 17: Cost impact for EURNOK/EURSEK. The figure shows how transaction costs affect strategy performance for the baseline, AR, and Markov-switching AR strategies, highlighting the sensitivity of cumulative returns to implementation frictions.

5.8.3 GBPUSD/EURUSD

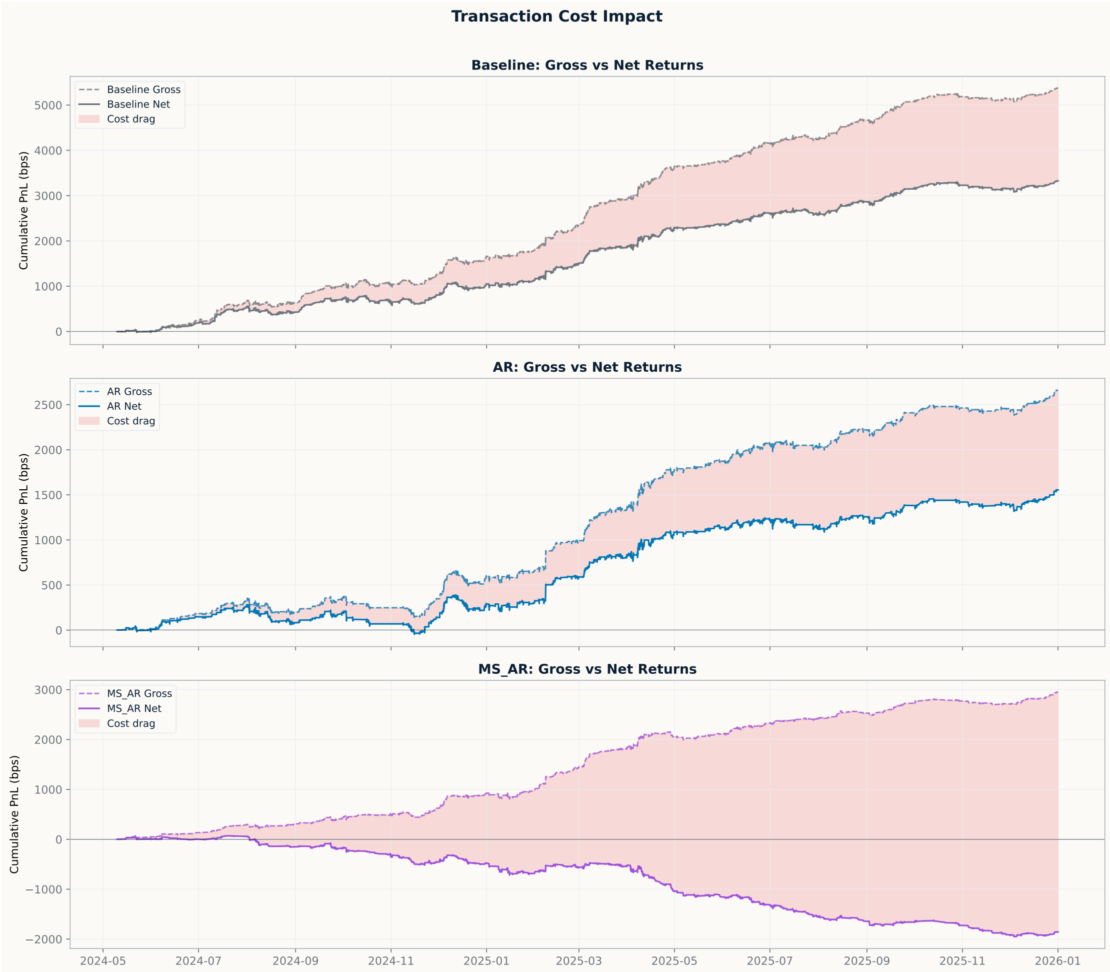

Figure 18: Cost impact for GBPUSD/EURUSD. The figure shows how transaction costs affect strategy performance for the baseline, AR, and Markov-switching AR strategies, highlighting the sensitivity of cumulative returns to implementation frictions.

5.9 Daily positions and regimes

5.9.1 AUDUSD/NZDUSD

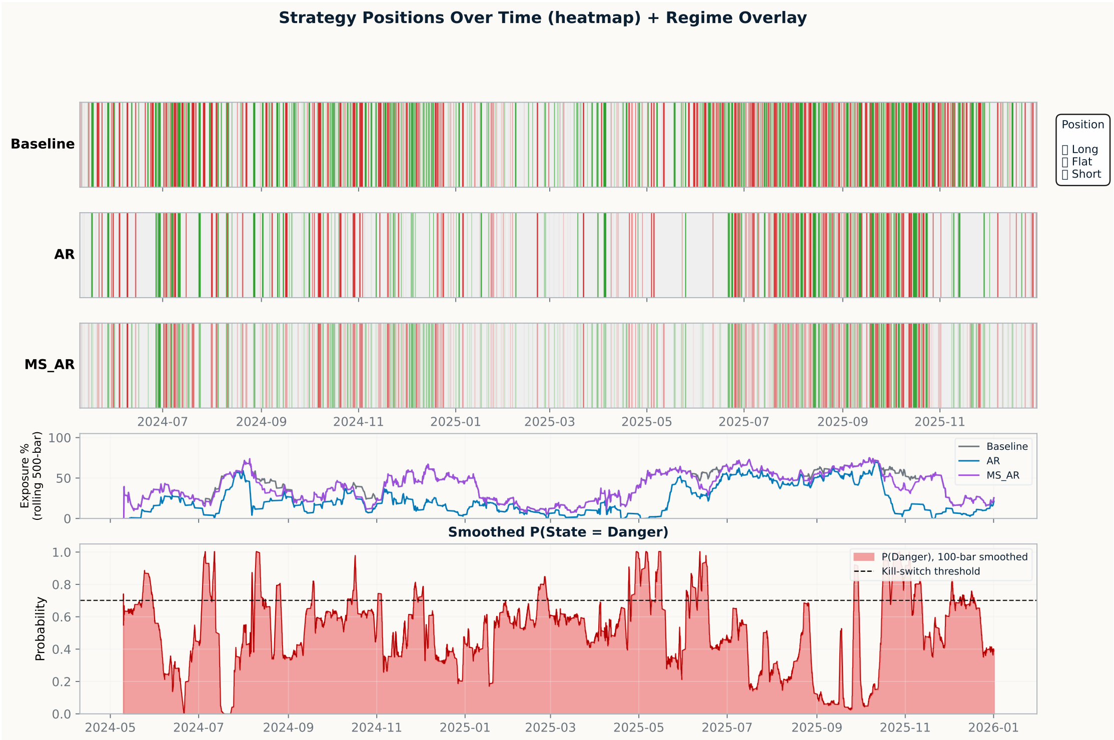

Figure 19: Daily positions and regime overlays for AUDUSD/NZDUSD. The figure shows daily trading exposure across the baseline, AR, and Markov-switching AR strategies, together with the smoothed dangerous-regime probability and the resulting long, flat, and short trading states.

5.9.2 EURNOK/EURSEK

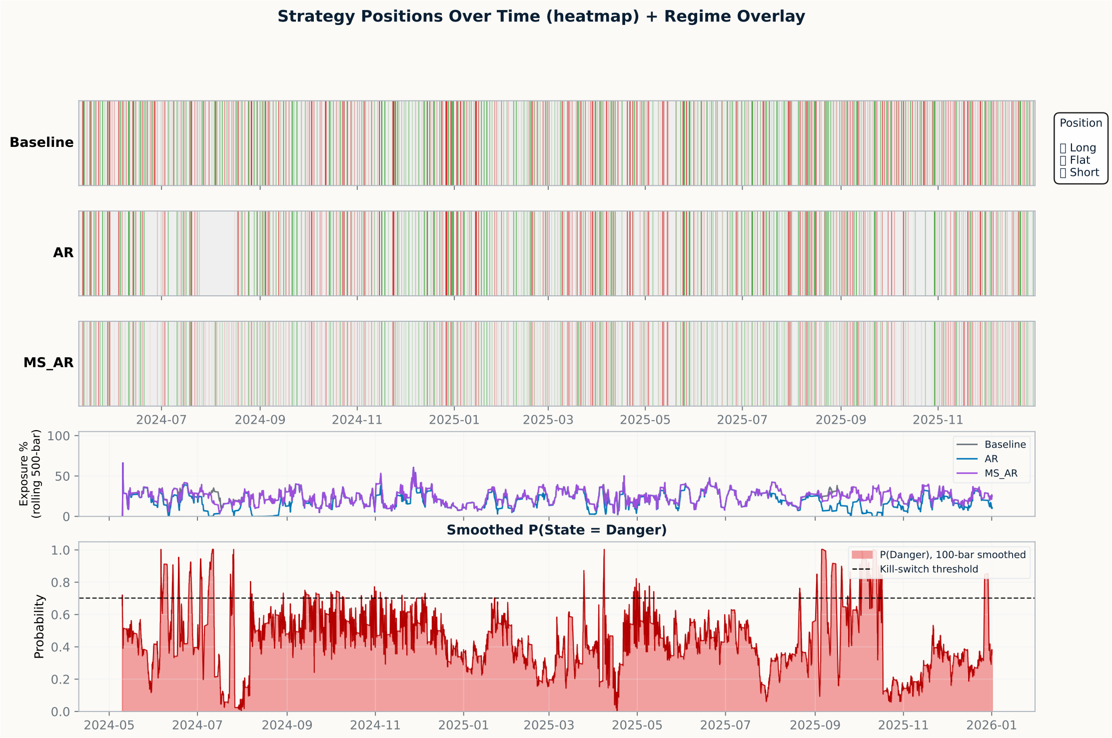

Figure 20: Daily positions and regime overlays for EURNOK/EURSEK. The figure shows daily trading exposure across the baseline, AR, and Markov-switching AR strategies, together with the smoothed dangerous-regime probability and the resulting long, flat, and short trading states.

5.9.3 GBPUSD/EURUSD

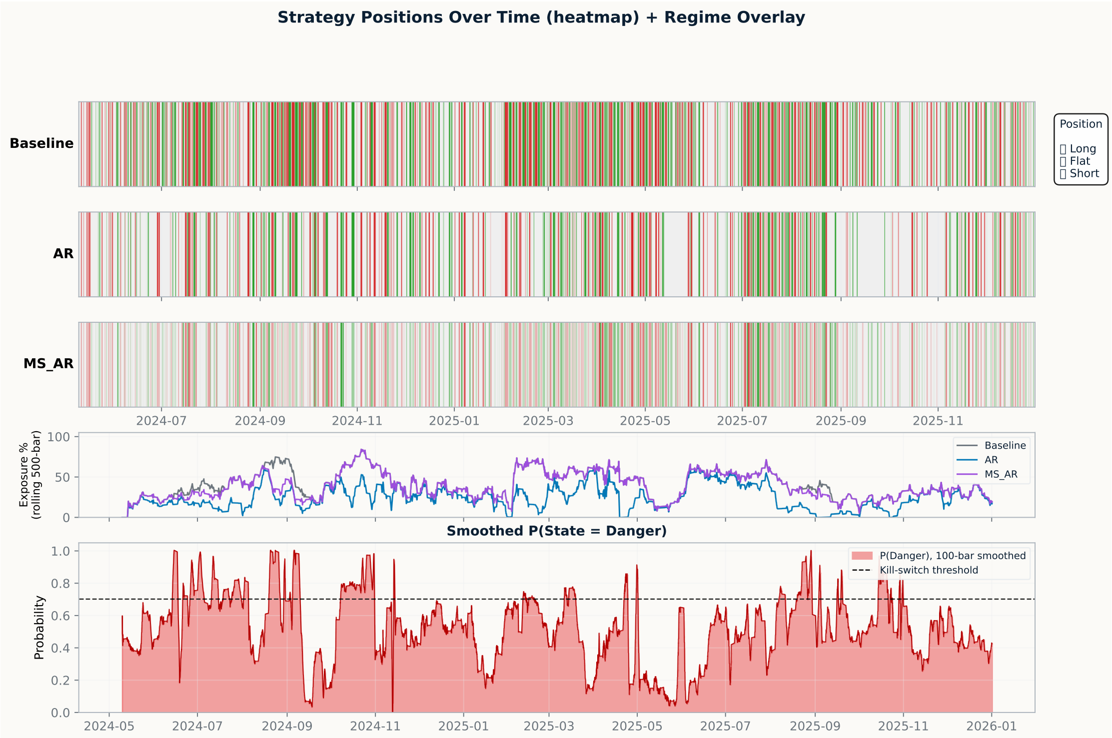

Figure 21: Daily positions and regime overlays for GBPUSD/EURUSD. The figure shows daily trading exposure across the baseline, AR, and Markov-switching AR strategies, together with the smoothed dangerous-regime probability and the resulting long, flat, and short trading states.

5.10 Daily backtest performance

5.10.1 AUDUSD/NZDUSD

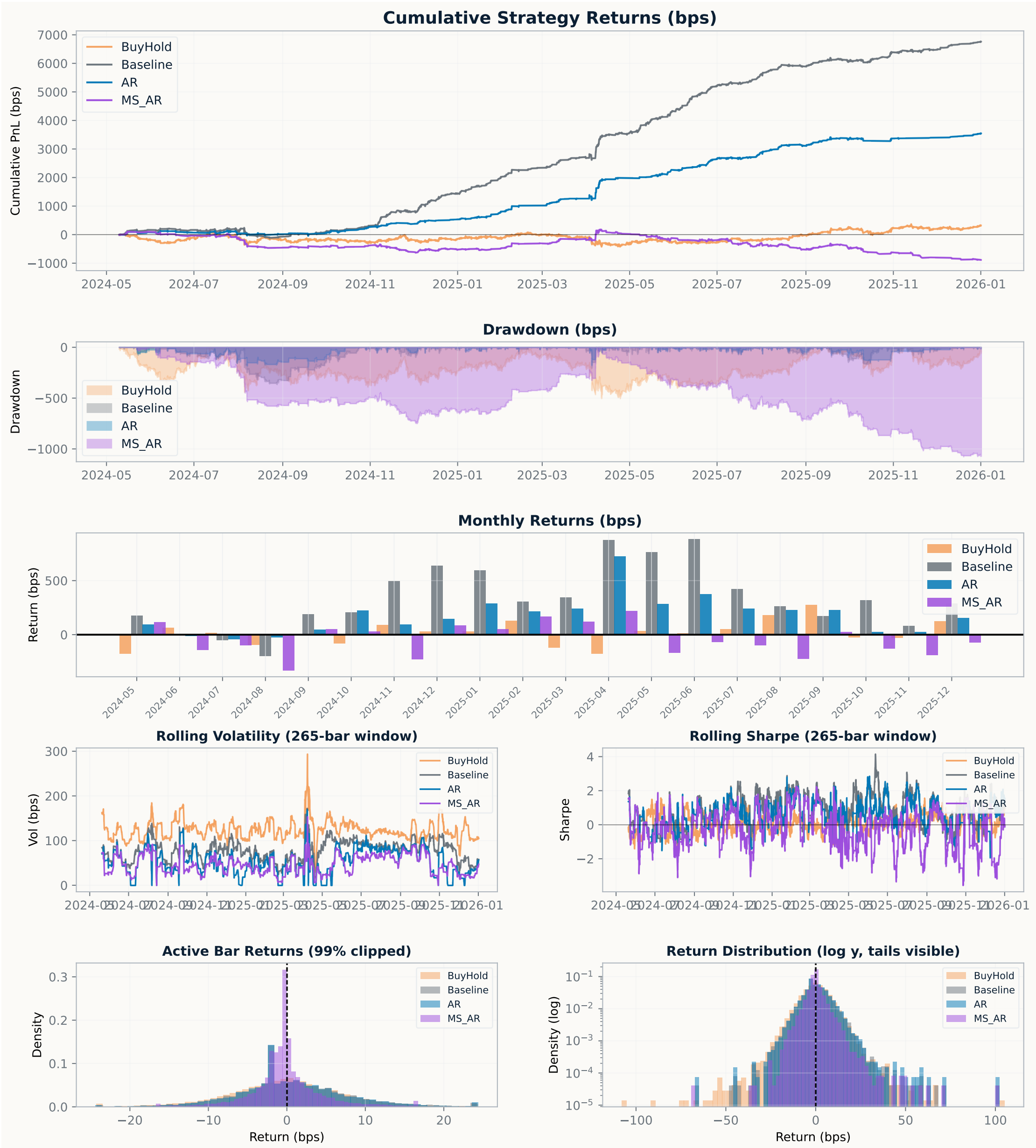

Figure 22: Daily backtest performance for AUDUSD/NZDUSD across the baseline, AR, and Markov-switching AR strategies. The figure compares cumulative strategy returns, drawdowns, monthly returns, rolling volatility, rolling Sharpe ratios, active-day return distributions, and tail behaviour on a log-density scale.

5.10.2 EURNOK/EURSEK

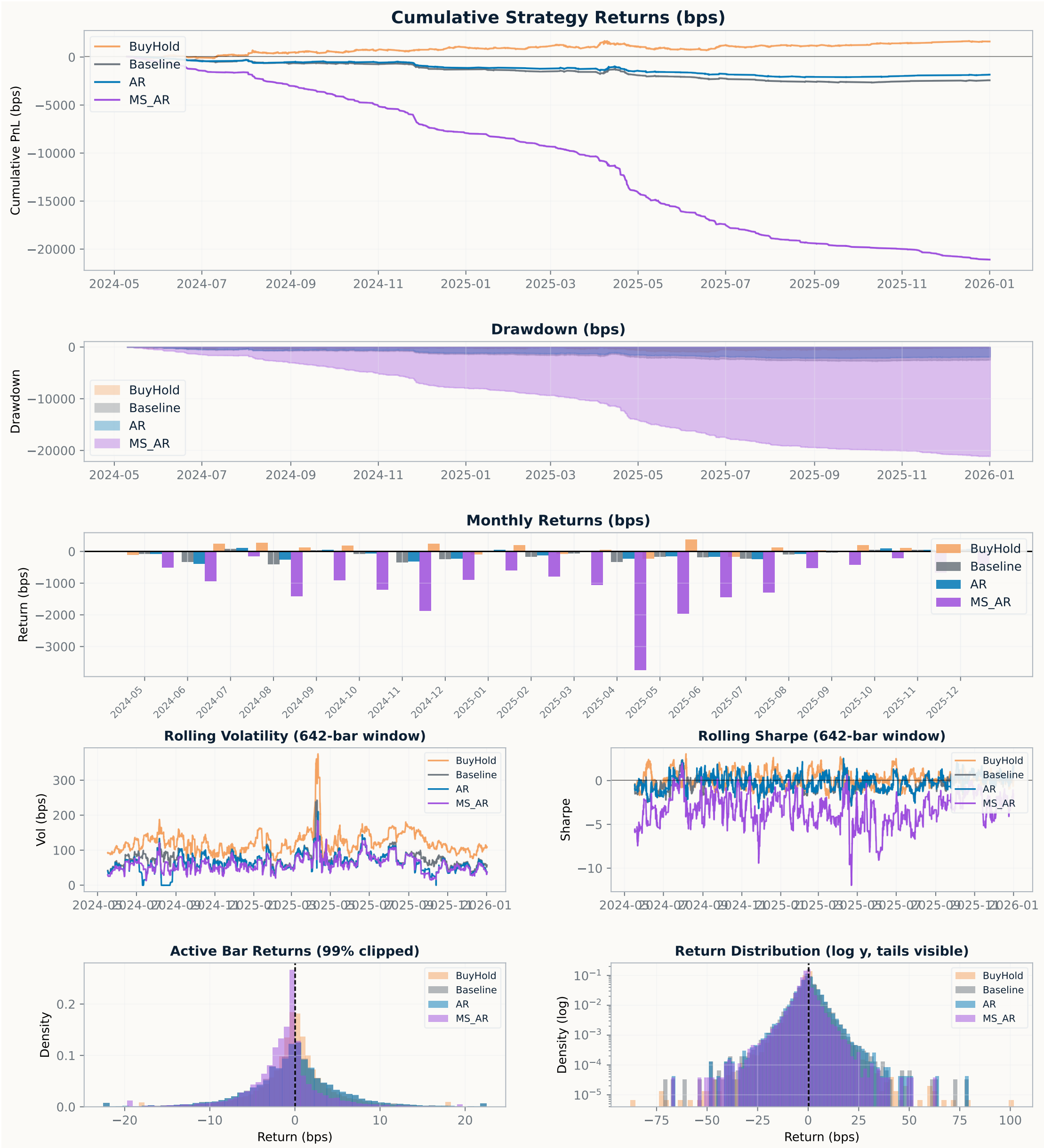

Figure 23: Daily backtest performance for EURNOK/EURSEK across the baseline, AR, and Markov-switching AR strategies. The figure compares cumulative returns, drawdowns, monthly return contributions, rolling volatility, rolling Sharpe ratios, active-day return densities, and tail behaviour.

5.10.3 GBPUSD/EURUSD

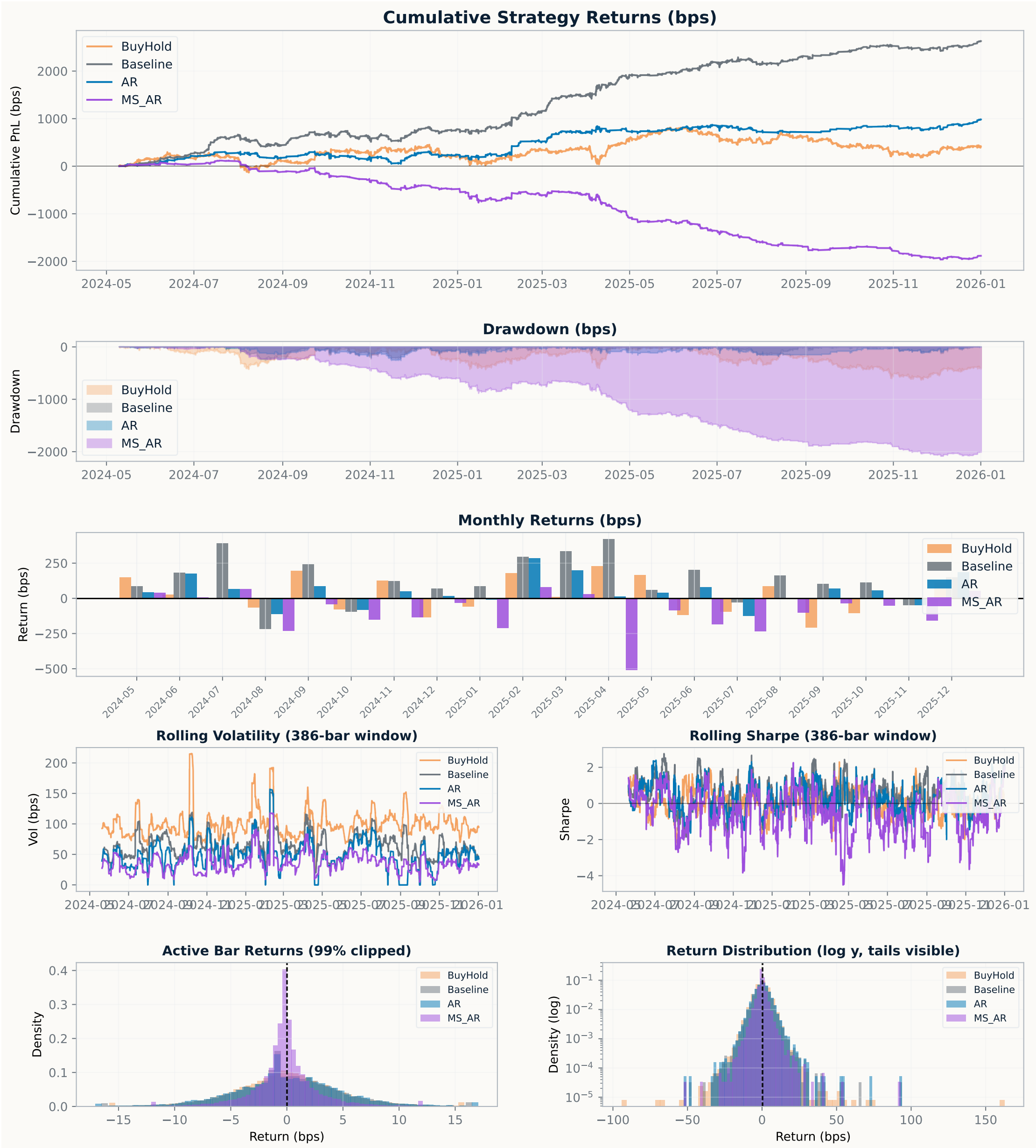

Figure 24: Daily backtest performance for GBPUSD/EURUSD across the baseline, AR, and Markov-switching AR strategies. The figure compares cumulative returns, drawdowns, monthly returns, rolling volatility, rolling Sharpe ratios, active-day return densities, and tail behaviour under the competing model specifications.